# Data Challenge pour les SHS

Analyse de données et introduction aux méthodes d'apprentissage automatique

Bénédicte Colnet - Julie Josse - Gael Varoquaux. (CP: JJ, GV, JP Nadal, Annick Vignes)

Lundi 25 janvier 2021

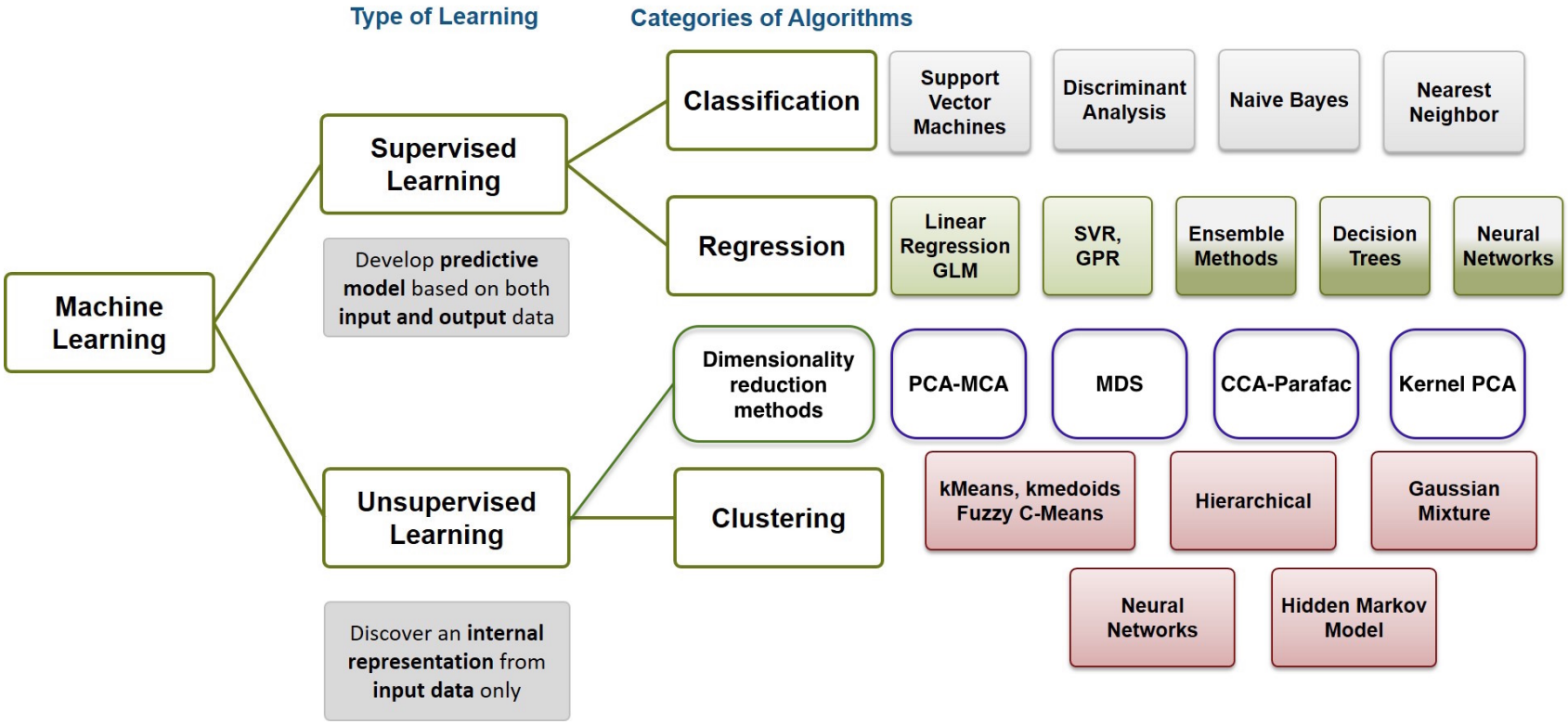
# Focus

- Applied statistics/ML and data analysis
- Apply and code a range of methods
- hands-on experience

$\Rightarrow$ Objectives:

- Perform a complete data analysis, from data manipulation, exploration, to analysis, to synthesis, to communication
- Method development to answer scientific questions
- Develop critical evaluation

# Machine Learning i




Supervised: Predict an outcome $Y$ from covariates $X_1, ..., X_p$

Unsupervised: No outcome

# Introduction

# Dimensionality reduction methods

$\Rightarrow$ Find a low-dimensional representation that captures the "essence" of high-dimensional data

- compression, denoising, data completion, anomaly detection
- preprocessing before supervised learning (improve performances / regularization to reduce overfitting)

$\Rightarrow$ descriptive methods, data visualization tools to better understand the data (difficult to plot and interpret $> 3d$)

- Principal component Analysis (PCA): continuous data
- Correspondence analysis (CA): contingency table
- Multiple correspondence analysis (MCA): categorical data
- Multiple factor analysis (MFA): multi-table, array data

Exploratory: "Let the data speak" J.P. Benzecri 1960.

`https://husson.github.io/MOOC_AnaDo/index.html`

# Principal Component Analysis

1. Data - Issues - Preprocessing
2. Observations Study
3. Variables Study
4. Interpretation Tools

# Data - Issues

# PCA for which data?

PCA deals with continuous variables, but categorical ones can also be included

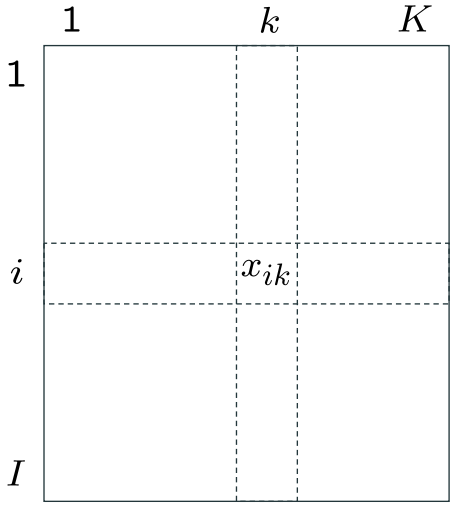



**Figure 1:** Data table

Many examples:

- Environment: waters - physico-chemical analyses
- Economy: countries - economic indicators
- Biology: tumors - genes expression
- Health: patients - clinical data
- Sociology: jobs- time spent for an activity
- etc.

# Wine data

- 10 observations (rows): white wines from Val de Loire
- 30 variables (columns):
  - 27 continuous variables: sensory descriptors
  - 2 continuous variables: odour and overall preferences
  - 1 categorical variable: label of the wines (Vouvray - Sauvignon)

| | O.fruity | O.passion | O.citrus | ... | Sweetness | Acidity | Bitterness | Astringency | Aroma.intensity | Aroma.persistency | Visual.intensity | Odor.preference | Overall.preference | Label |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| S Michaud | 4,3 | 2,4 | 5,7 | ... | 3,5 | 5,9 | 4,1 | 1,4 | 7,1 | 6,7 | 5,0 | 6,0 | 5,0 | Sauvignon |
| S Renaudie | 4,4 | 3,1 | 5,3 | ... | 3,3 | 6,8 | 3,8 | 2,3 | 7,2 | 6,6 | 3,4 | 5,4 | 5,5 | Sauvignon |
| S Trotignon | 5,1 | 4,0 | 5,3 | ... | 3,0 | 6,1 | 4,1 | 2,4 | 6,1 | 6,1 | 3,0 | 5,0 | 5,5 | Sauvignon |
| S Buisse Domaine | 4,3 | 2,4 | 3,6 | ... | 3,9 | 5,6 | 2,5 | 3,0 | 4,9 | 5,1 | 4,1 | 5,3 | 4,6 | Sauvignon |
| S Buisse Cristal | 5,6 | 3,1 | 3,5 | ... | 3,4 | 6,6 | 5,0 | 3,1 | 6,1 | 5,1 | 3,6 | 6,1 | 5,0 | Sauvignon |
| V Aub Silex | 3,9 | 0,7 | 3,3 | ... | 7,9 | 4,4 | 3,0 | 2,4 | 5,9 | 5,6 | 4,0 | 5,0 | 5,5 | Vouvray |
| V Aub Marigny | 2,1 | 0,7 | 1,0 | ... | 3,5 | 6,4 | 5,0 | 4,0 | 6,3 | 6,7 | 6,0 | 5,1 | 4,1 | Vouvray |
| V Font Domaine | 5,1 | 0,5 | 2,5 | ... | 3,0 | 5,7 | 4,0 | 2,5 | 6,7 | 6,3 | 6,4 | 4,4 | 5,1 | Vouvray |
| V Font Brûlés | 5,1 | 0,8 | 3,8 | ... | 3,9 | 5,4 | 4,0 | 3,1 | 7,0 | 6,1 | 7,4 | 4,4 | 6,4 | Vouvray |
| V Font Coteaux | 4,1 | 0,9 | 2,7 | ... | 3,8 | 5,1 | 4,3 | 4,3 | 7,3 | 6,6 | 6,3 | 6,0 | 5,7 | Vouvray |

# Degustation

# Degustation

# Degustation

# Objectives

- Observations study:
  similarity between observations with respect to all the variables
  partition between observations

- Variables study:
  linear relationships between variables
  visualization of the correlation matrix
  find synthetic variables

- Link between the two studies:
  characterization of the groups of observations with variables
  specific observations to understand links between variables

# Observations study

# Preprocessing... (not an appropriate word?)

$\Rightarrow$ Similarity between observations: Euclidean distance

- Choosing active variables

$$d^2(i,i') = \sum_{k=1}^{K}(x_{ik} - x_{i'k})^2$$

- Variables are centered

$$d^2(i,i') = \sum_{k=1}^{K}((x_{ik} - \bar{x}_k - (x_{i'k} - \bar{x}_k))^2$$

$x_{ik} \hookrightarrow x_{ik} - \bar{x}_k$

- Standardizing variables or not?

$$d^2(i,i') = \sum_{k=1}^{K}\frac{1}{s_k^2}(x_{ik} - x_{i'k})^2$$

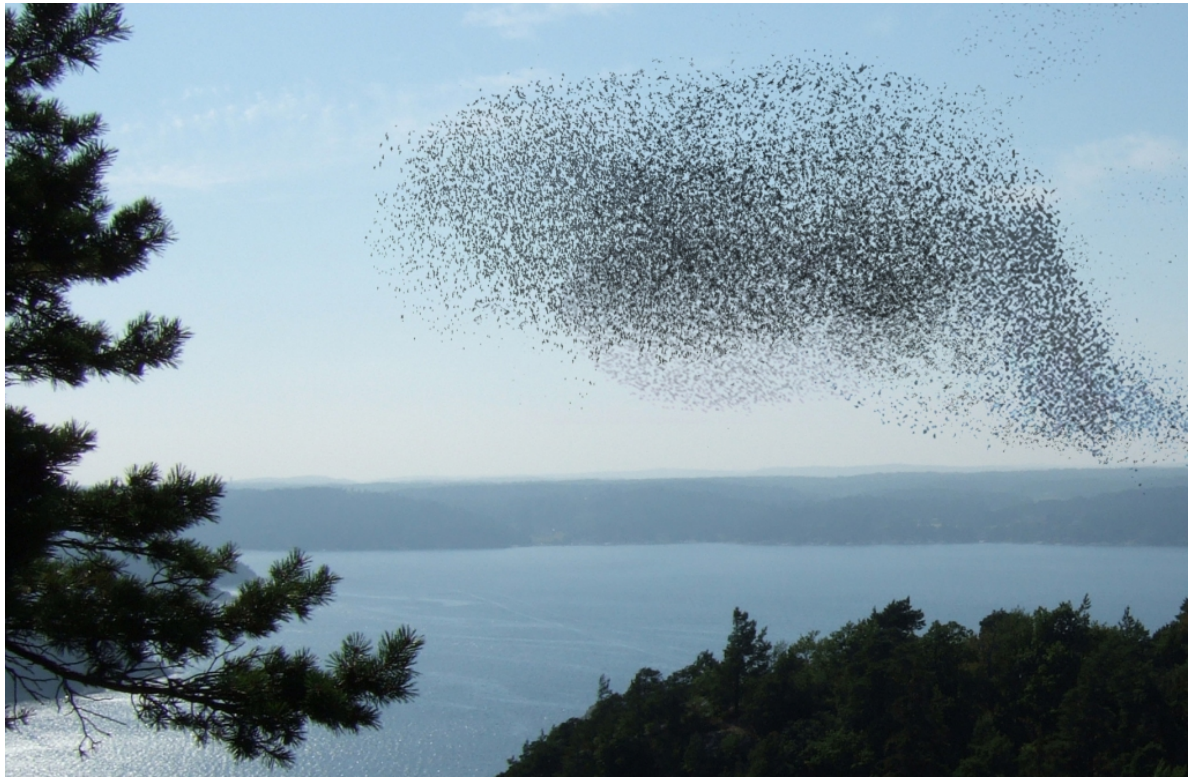
$\bar{x}_k = \frac{1}{I}\sum_{i=1}^{I} x_{ik}$ and $s_k^2 = \frac{1}{n}\sum_{i=1}^{I}(x_{ik} - \bar{x}_k)$

# Wine data

- Sensory descriptors as active: only these variables are used to define the dimensions
- Variables are (centered and) standardized

| | O.fruity | O.passion | O.citrus | ... | Sweetness | Acidity | Bitterness | Astringency | Aroma.intensity | Aroma.persistency | Visual.intensity | Odor.preference | Overall.preference | Label |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| S Michaud | 4,3 | 2,4 | 5,7 | ... | 3,5 | 5,9 | 4,1 | 1,4 | 7,1 | 6,7 | 5,0 | 6,0 | 5,0 | Sauvignon |
| S Renaudie | 4,4 | 3,1 | 5,3 | ... | 3,3 | 6,8 | 3,8 | 2,3 | 7,2 | 6,6 | 3,4 | 5,4 | 5,5 | Sauvignon |
| S Trotignon | 5,1 | 4,0 | 5,3 | ... | 3,0 | 6,1 | 4,1 | 2,4 | 6,1 | 6,1 | 3,0 | 5,0 | 5,5 | Sauvignon |
| S Buisse Domaine | 4,3 | 2,4 | 3,6 | ... | 3,9 | 5,6 | 2,5 | 3,0 | 4,9 | 5,1 | 4,1 | 5,3 | 4,6 | Sauvignon |
| S Buisse Cristal | 5,6 | 3,1 | 3,5 | ... | 3,4 | 6,6 | 5,0 | 3,1 | 6,1 | 5,1 | 3,6 | 6,1 | 5,0 | Sauvignon |
| V Aub Silex | 3,9 | 0,7 | 3,3 | ... | 7,9 | 4,4 | 3,0 | 2,4 | 5,9 | 5,6 | 4,0 | 5,0 | 5,5 | Vouvray |
| V Aub Marigny | 2,1 | 0,7 | 1,0 | ... | 3,5 | 6,4 | 5,0 | 4,0 | 6,3 | 6,7 | 6,0 | 5,1 | 4,1 | Vouvray |
| V Font Domaine | 5,1 | 0,5 | 2,5 | ... | 3,0 | 5,7 | 4,0 | 2,5 | 6,7 | 6,3 | 6,4 | 4,4 | 5,1 | Vouvray |
| V Font Brûlés | 5,1 | 0,8 | 3,8 | ... | 3,9 | 5,4 | 4,0 | 3,1 | 7,0 | 6,1 | 7,4 | 4,4 | 6,4 | Vouvray |
| V Font Coteaux | 4,1 | 0,9 | 2,7 | ... | 3,8 | 5,1 | 4,3 | 4,3 | 7,3 | 6,6 | 6,3 | 6,0 | 5,7 | Vouvray |

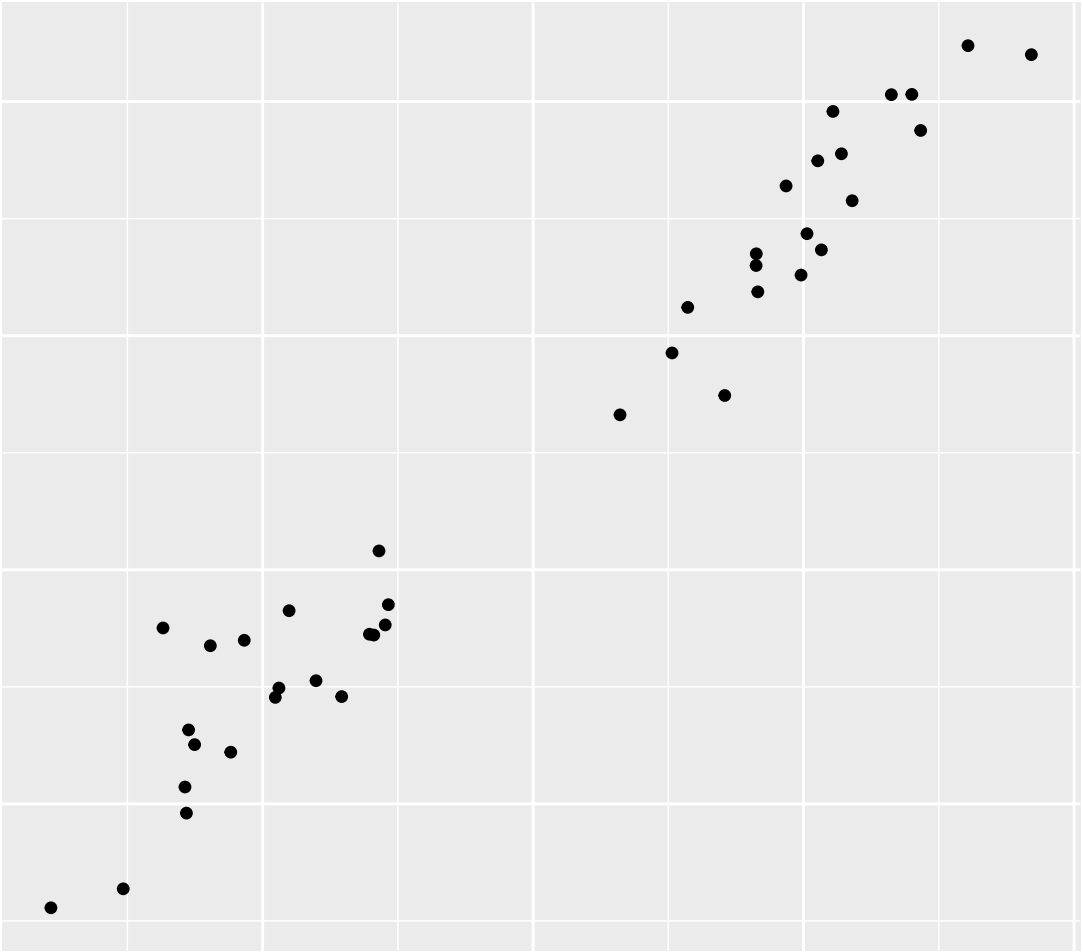
# Observations cloud



- Study the structure, *i.e.* the shape of the cloud of observations
- Observations are in $\mathbb{R}^K$

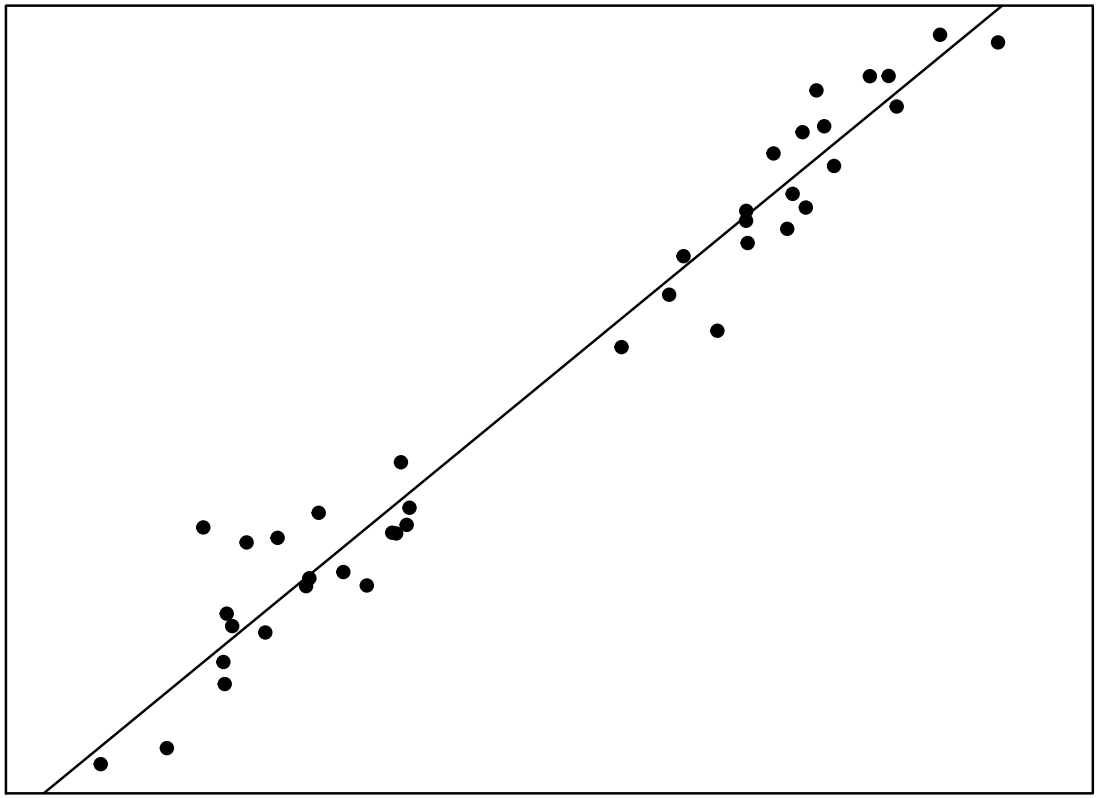
# Fit the observations cloud

Find the subspace which best sums up the data



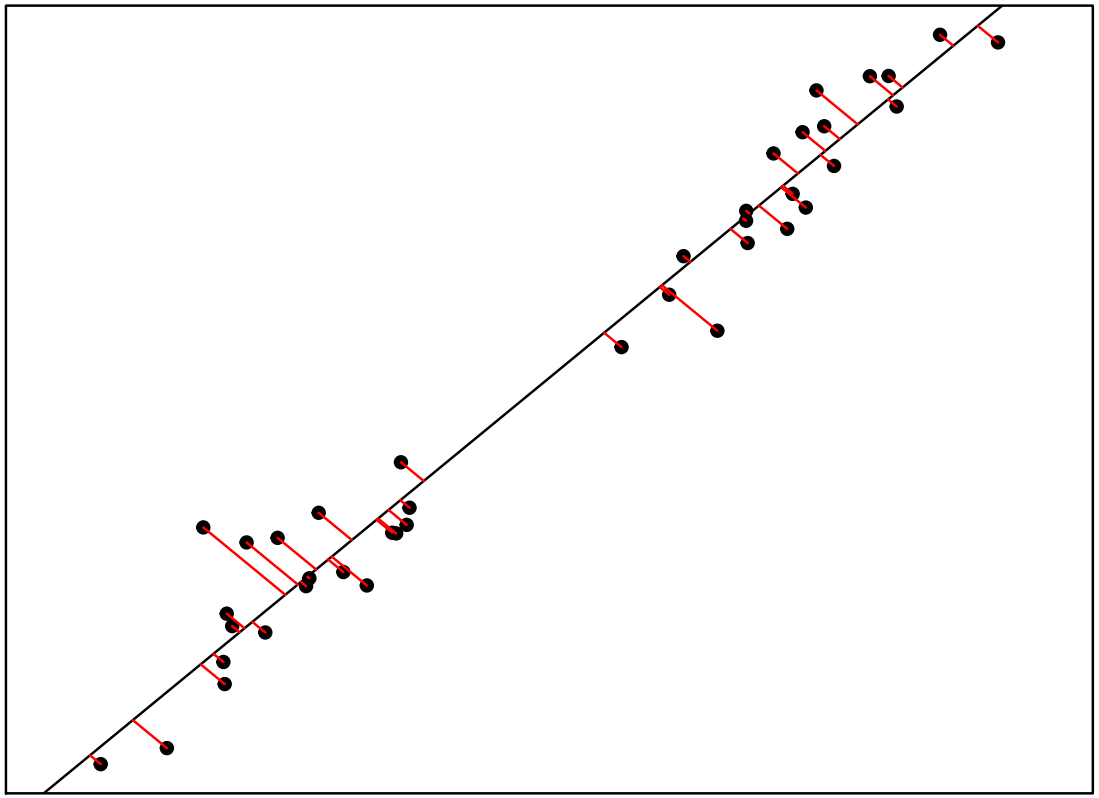
# Fit the observations cloud

Find the subspace which best sums up the data



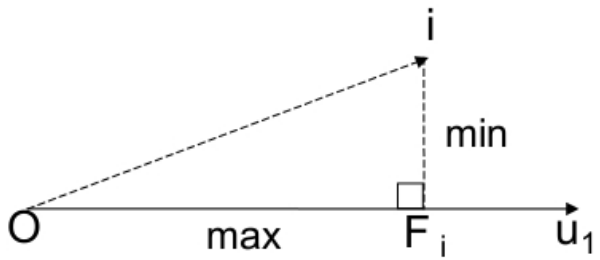
# Fit the observations cloud

Find the subspace which best sums up the data



# Fit the observations cloud

- Find the axis $u_1$ that deforms less the data.




$iF_{i1}$ small $\Leftrightarrow$ $OF_{i1}$ large

max $\sum_i (OF_{i1})^2$

$u_1^\star = \arg\max_{u_1 \in \mathbb{R}^K} \frac{1}{I} \sum_{i=1}^{I} F_{i1}^2$

$u_1$ loadings - $F_{.1}$ principal component (PC), scores

- Find the best subspace: look for $u_2 \perp u_1$ which maximizes the variance (inertia) of the projected points, max $\sum_i (OF_{i2})^2$ etc.

$\Rightarrow$ Representation quality (loss information):
Percentage of inertia explained: Variance of the projected obs cloud/ Total variance variance of the obs cloud
$Q$-dim representation $var(F_{.1}) + var(F_{.2}) + ... + var(F_{.Q})/$
$\sum_i \frac{1}{I} d(x_{i.}, G_I)^2 = \frac{1}{I} \sum_i \sum_k (x_{ik})^2 = \sum_{k=1}^{K} \lambda_k$

# Fit the observations cloud

Find the subspace which best sums up the data

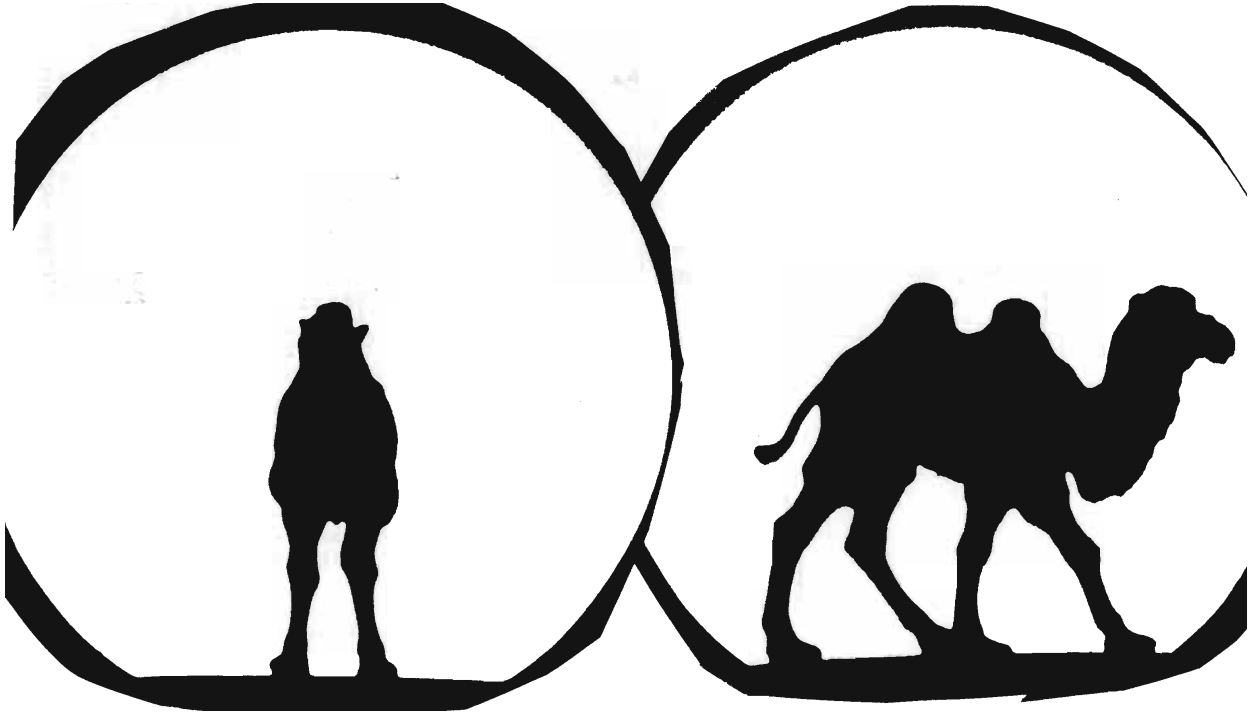
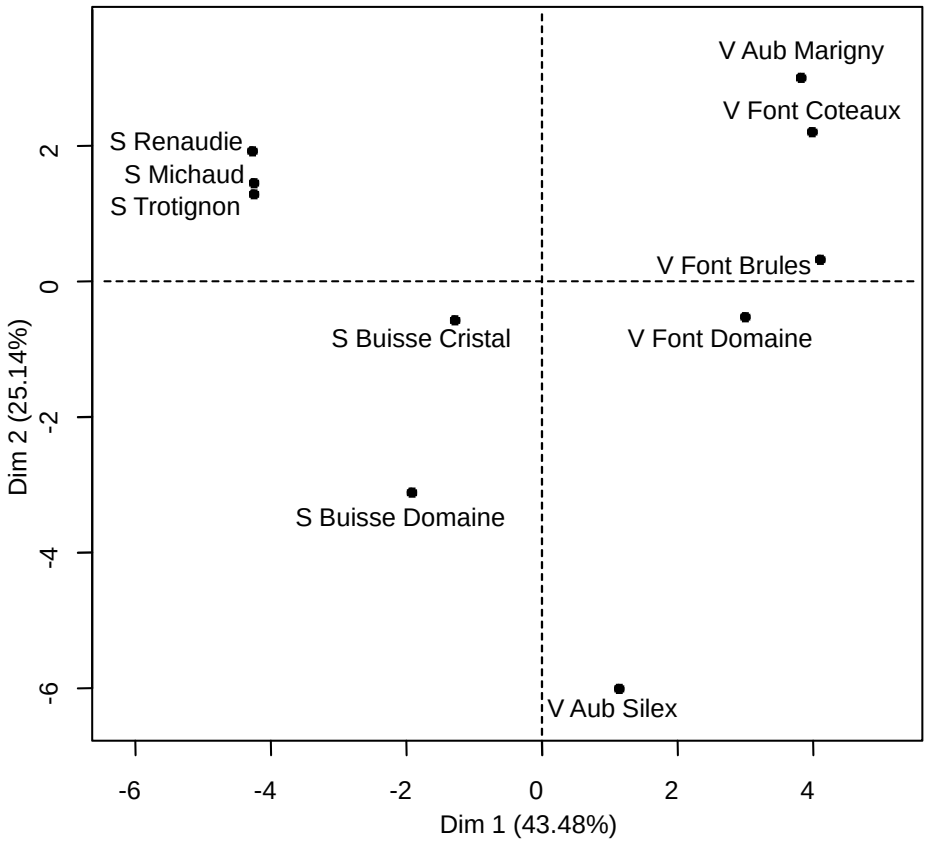


**Figure 2:** Camel vs dromedary? (J.P. Fenelon)

$\Rightarrow$ Closest representation by projection

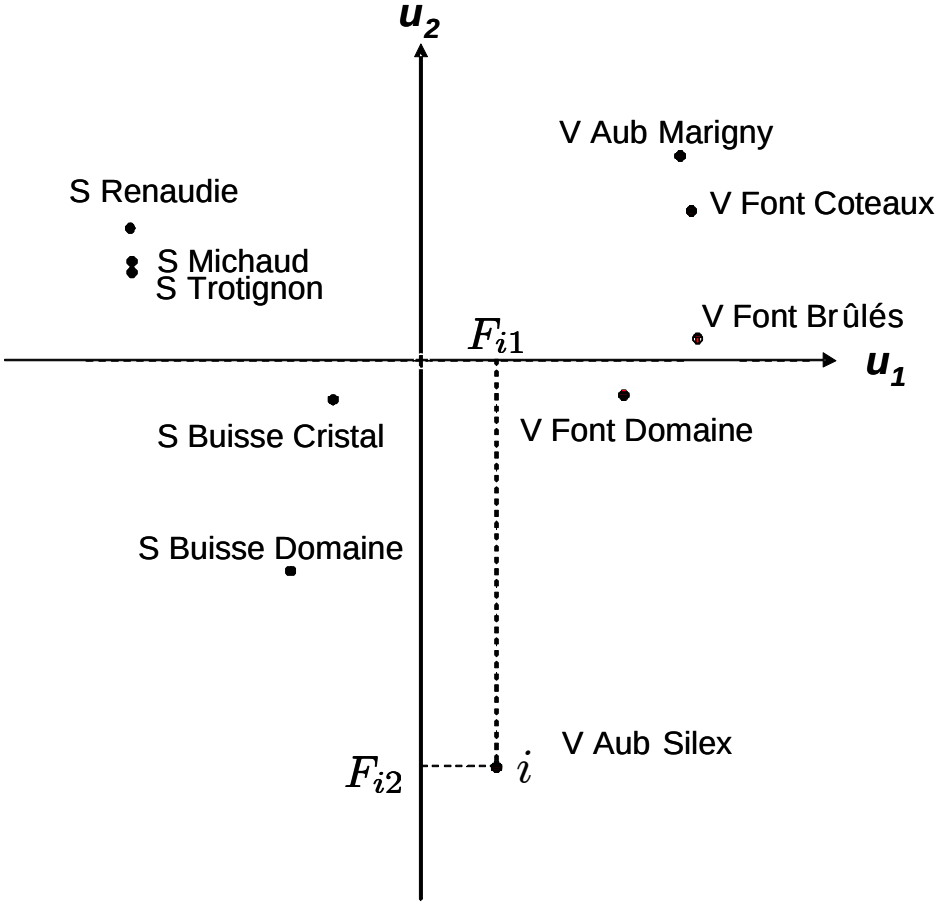
$\Rightarrow$ Best representation of the diversity, variability

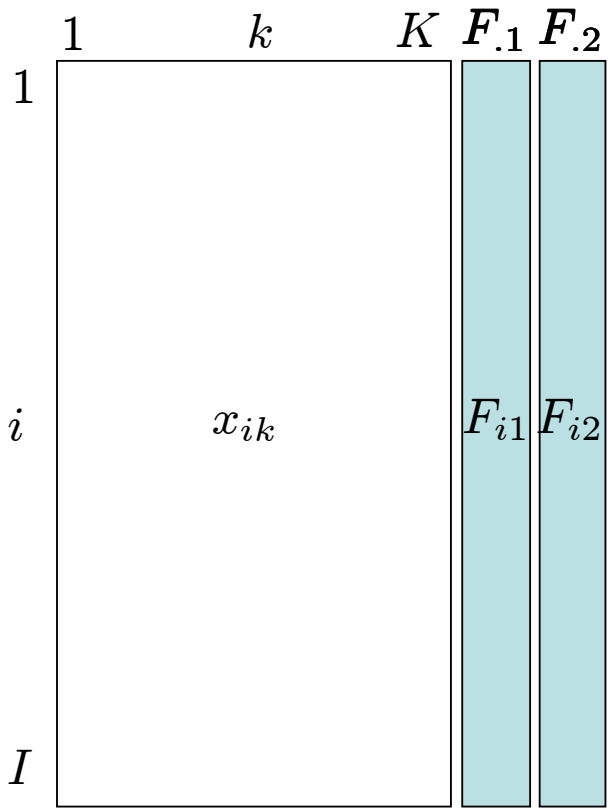
# Wine: graph of observations




$\Rightarrow$ Need variables to interpret the dimensions of variability

# Observations coordinates considered as variables

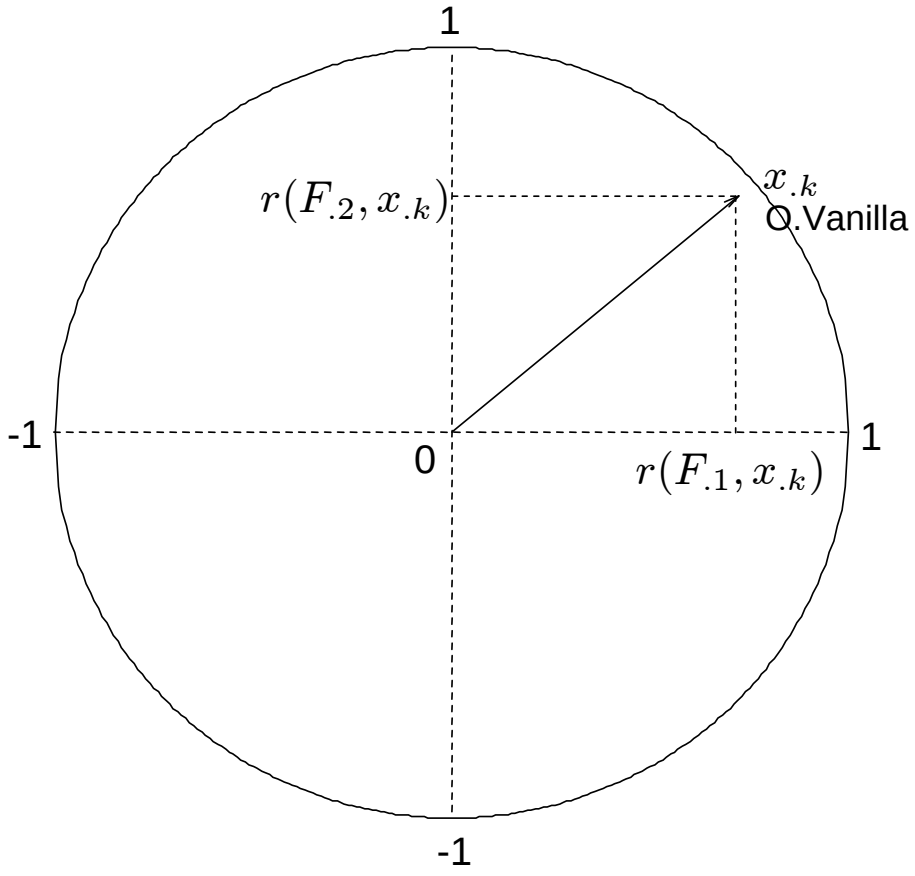



$F = Xu$ (linear combination of variables)

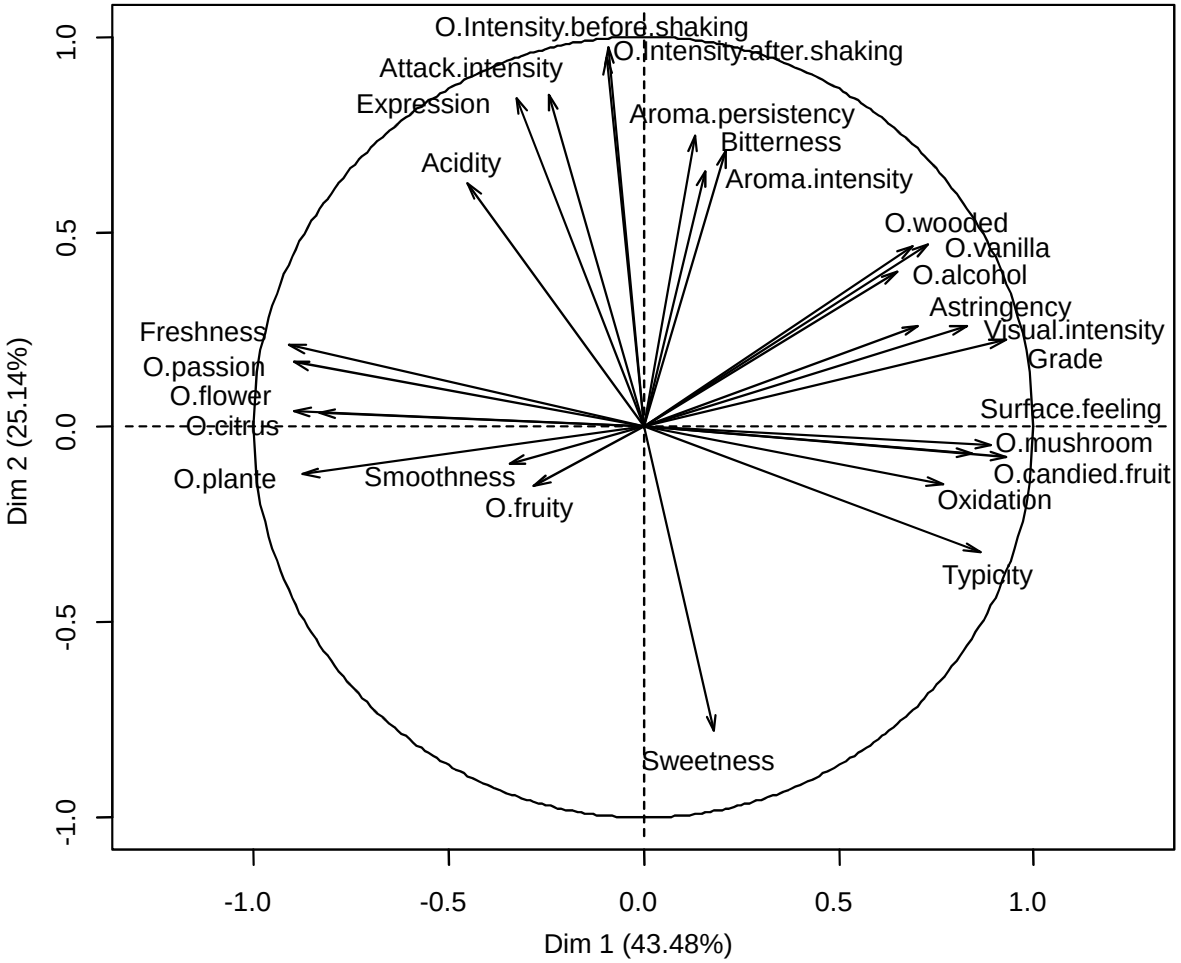
# Interpretation of the observations graph with the variables

- Correlation between variable $x_{.k}$ and $F_{.1}$ (and $F_{.2}$)



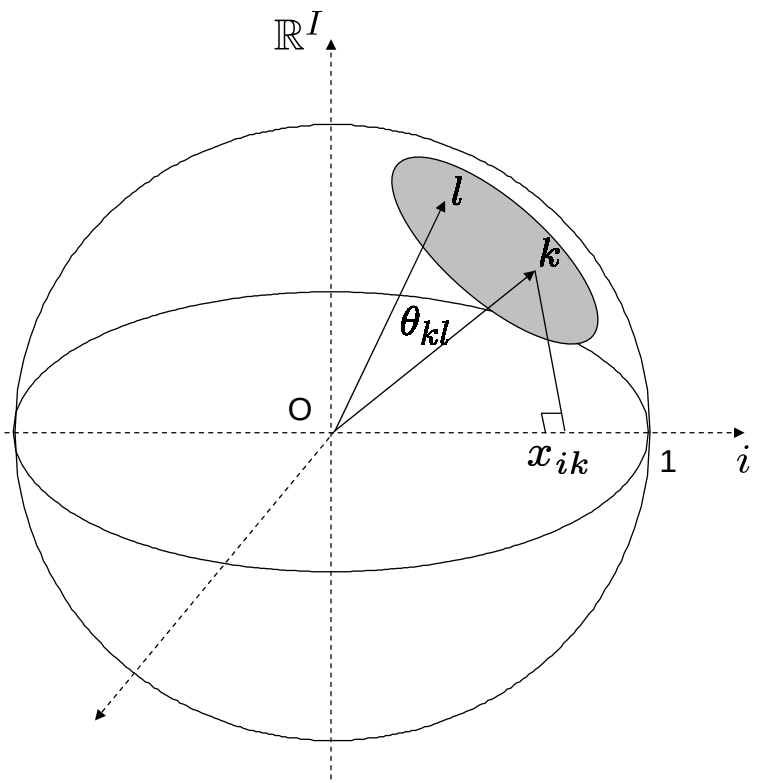

$\Rightarrow$ Correlation circle

# Interpretation of the observations graph with the variables



# Variables study

# Cloud of variables



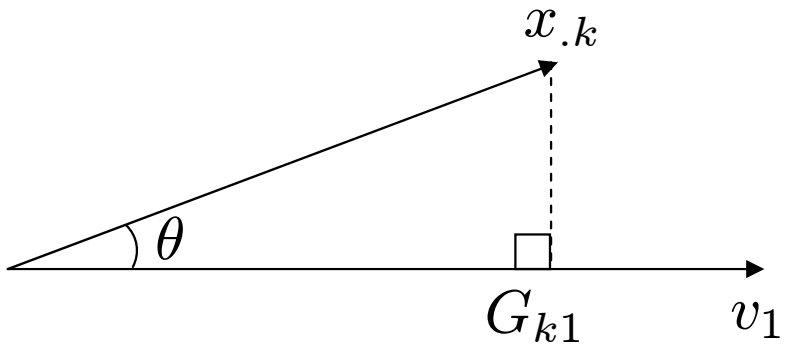

$$cos(\theta_{kl}) = \frac{< x_{.k}, x_{.l} >}{\|x_{.k}\| \; \|x_{.l}\|} = \frac{\sum_{i=1}^{I} x_{ik} x_{il}}{\sqrt{(\sum_{i=1}^{I} x_{ik}^2)(\sum_{i=1}^{I} x_{il}^2)}} = r(x_{.k}, x_{.l})$$

variables are centered: cosine = correlation.

variables are standardized: on an hypersphere of radius 1

# Fit the variables cloud

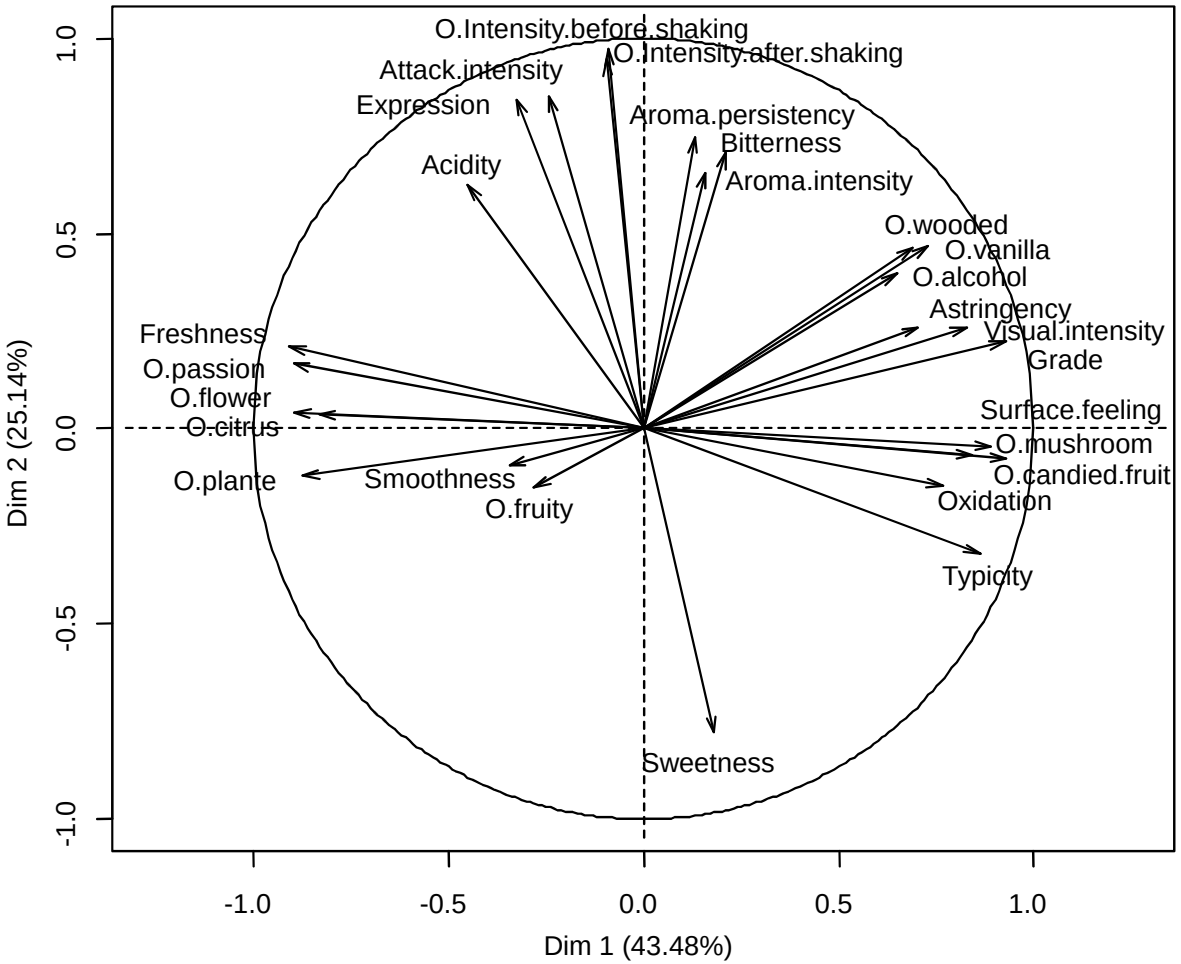
Find $v_1$ which best fits the cloud




$$\underset{v_1 \in \mathbb{R}^I}{\arg\max} \sum_{i=k}^{K} G_{k1}^2 = \underset{v_1 \in \mathbb{R}^I}{\arg\max} \sum_{i=k}^{K} r(v_1, x_{.k})^2$$

$\Rightarrow$ $v_1$ is the best synthetic variable

# Fit the variables cloud

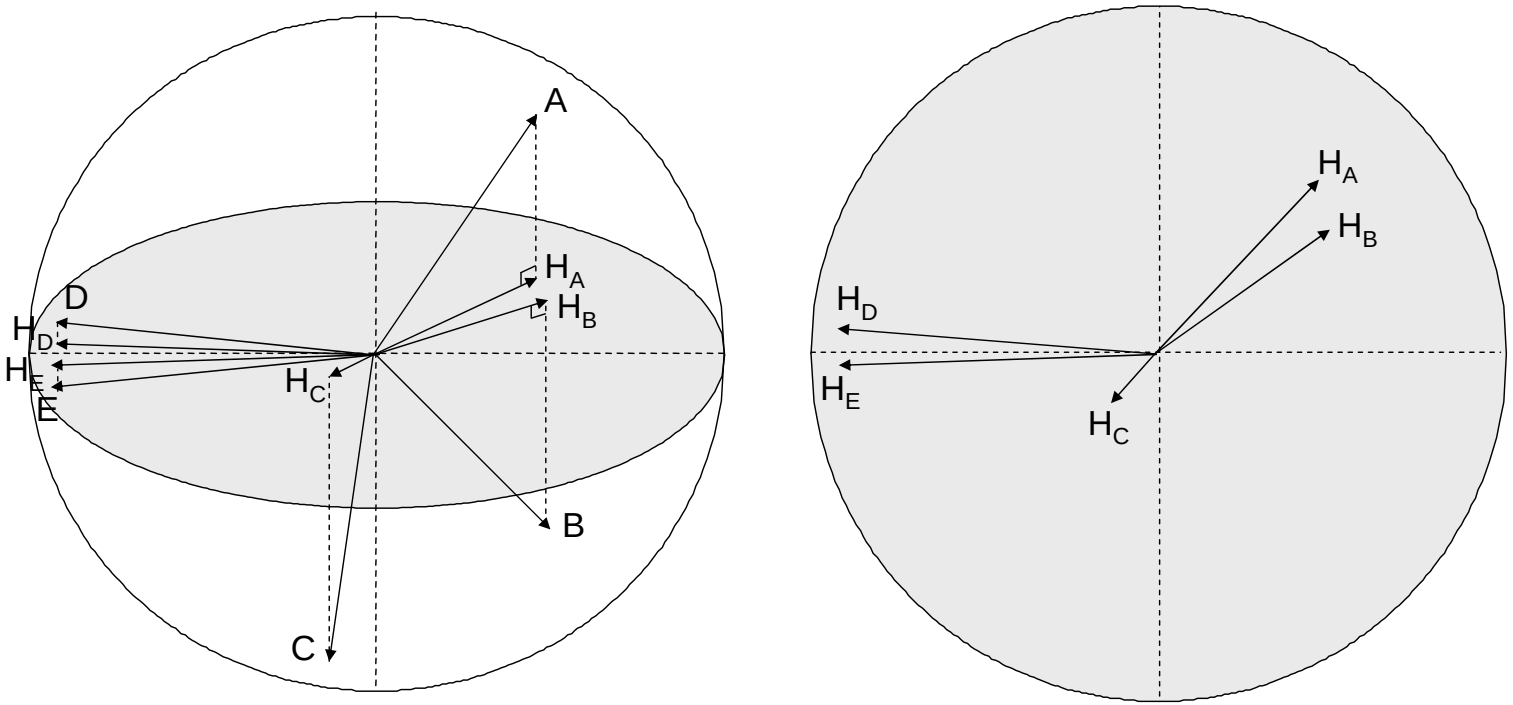



$\Rightarrow$ Same representation

# Projections...

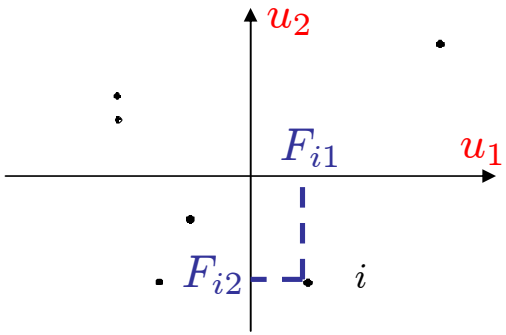
$r(x_{.1}, x_{.2}) = cos(\theta_{x_{.1},x_{.2}})$

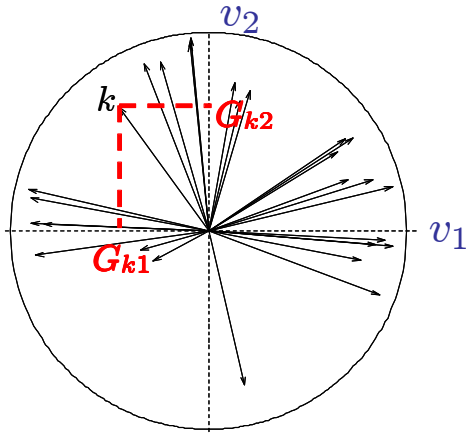
$cos(\theta_{x_{.1},x_{.2}}) \approx cos(\theta_{H_1,H_2})$ if variables are well projected

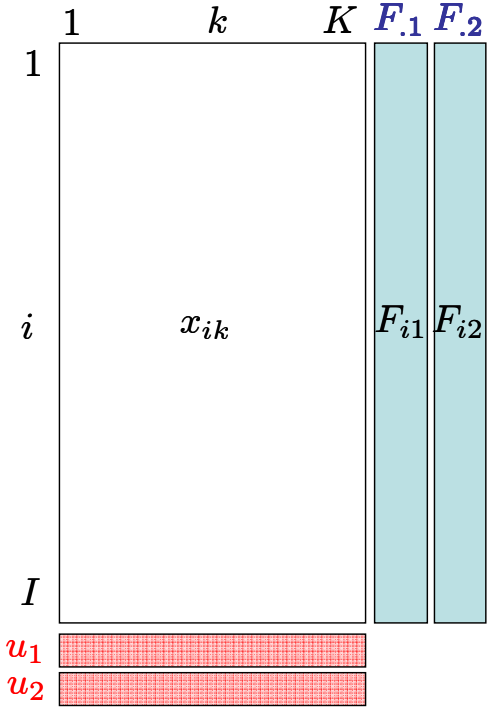


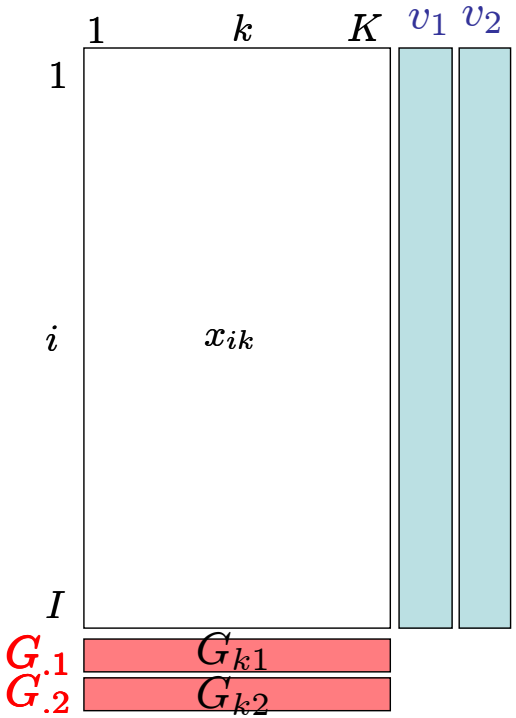

Only well projected variables can be interpreted

# Link between the two representations: transition formulae



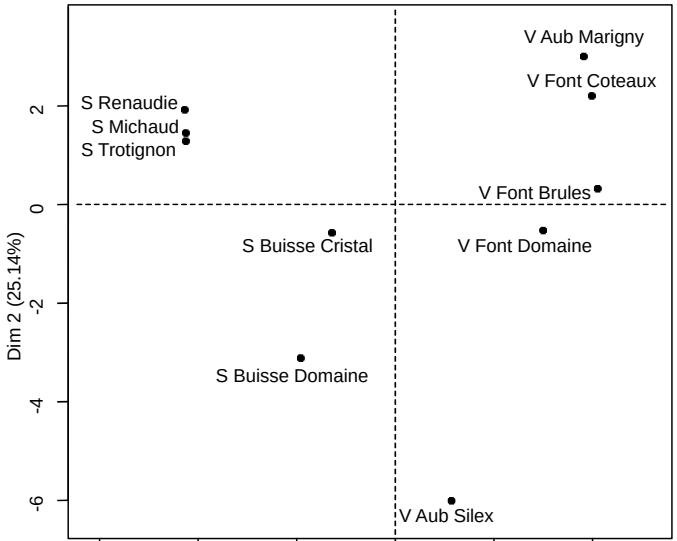



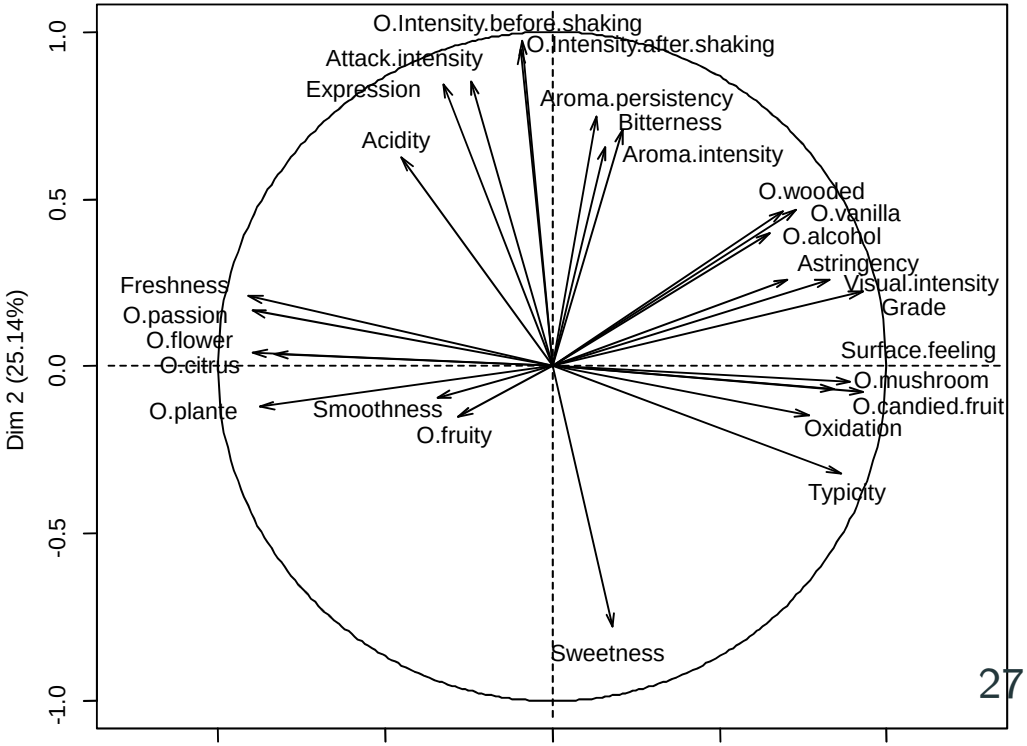
# Link between the two representations: transition formulae

$$F_{iq} = \frac{1}{\sqrt{\lambda_q}} \sum_{k=1}^{K} x_{ik} G_{kq}$$

$$G_{kq} = \frac{1}{\sqrt{\lambda_q}} \sum_{i=1}^{I} (1/I) x_{ik} F_{iq}$$

Observation on the side of the variables where it takes high values

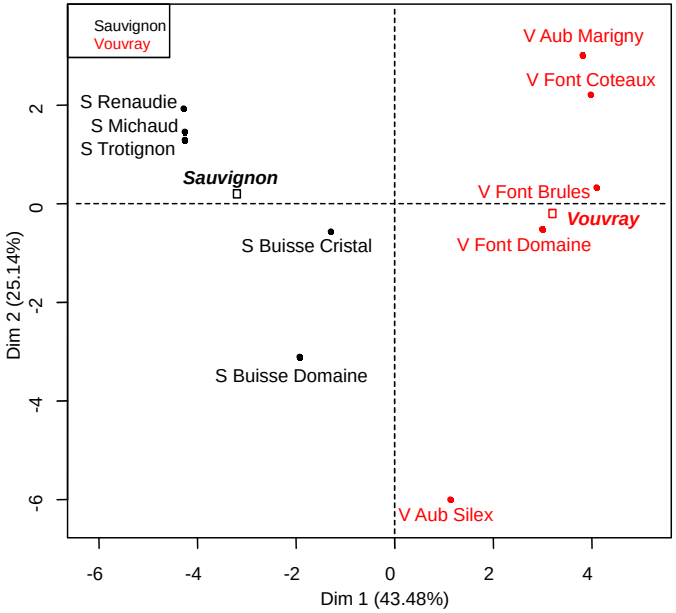


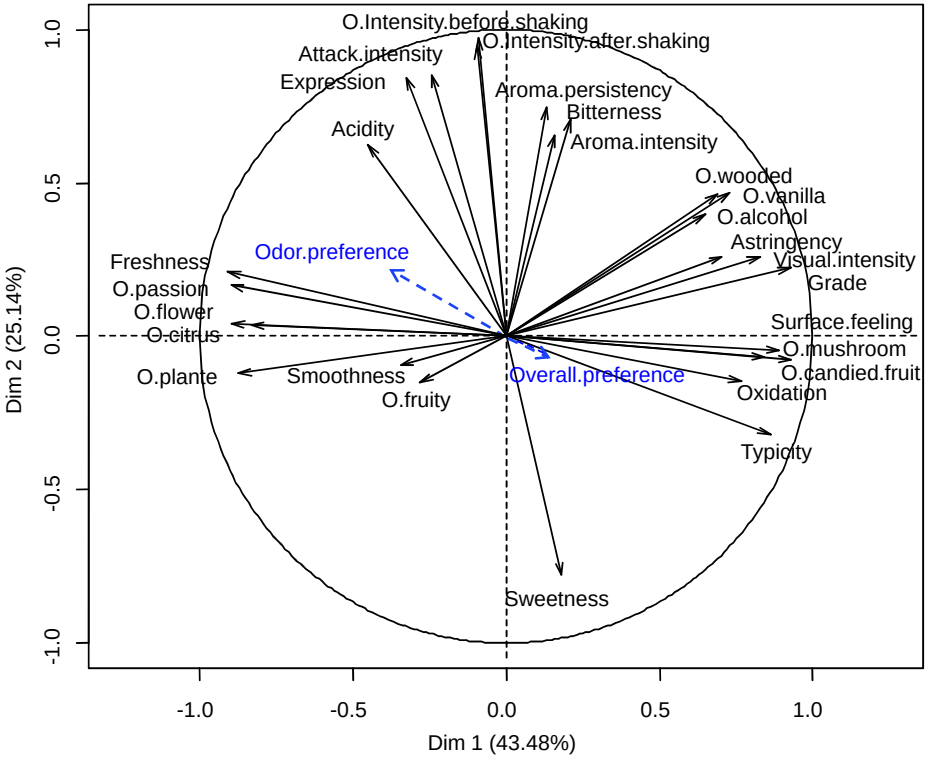



# Interpretation tools

# Supplementary information

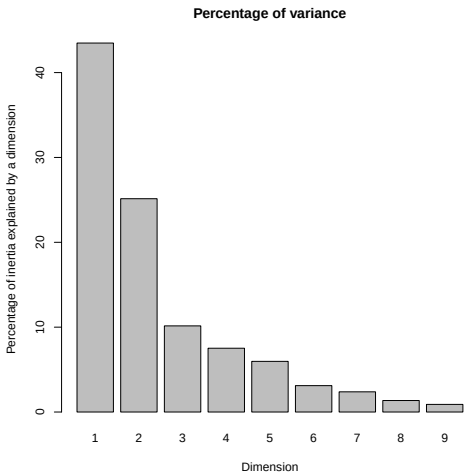
- continuous variables: projection (correlation with dimensions)
- observations: projection
- categorical variables: projection of the categories at the barycentre of the observations which take the categories



# Choosing the number of components

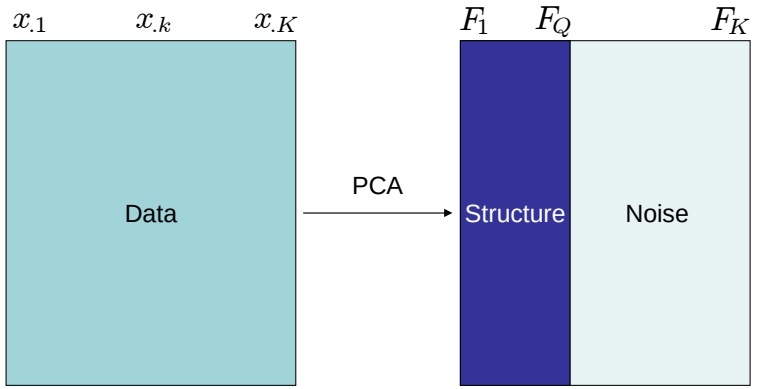
Bar plot, test on eigenvalues, confidence interval, cross-validation, generalized cross-validation, etc.




Objectives:

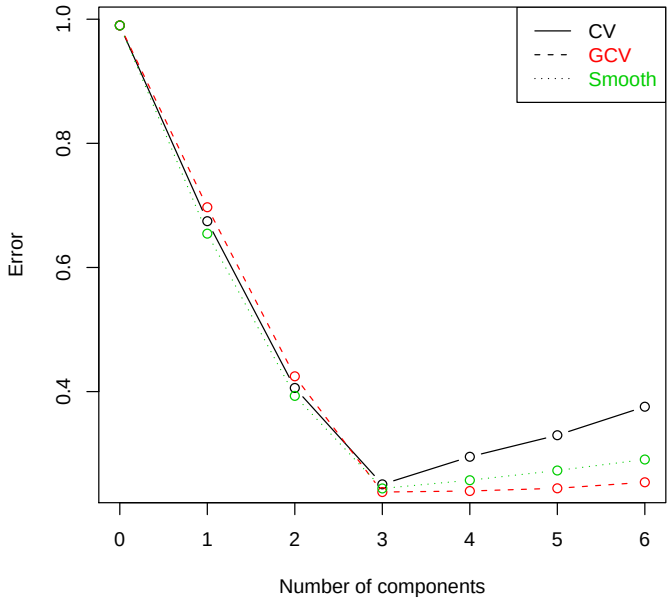
$\Rightarrow$ Interpretation

$\Rightarrow$ Separate structure and noise



# Approximations

```
> nb <- estim_ncp(don)
> nb$ncp
> nb$criterion
        0         1         2         3         4         5
1.2884873 0.8069719 0.6400517 0.7045074 2.2257738 3.0274337
```




Josse, J. & Husson, F. (2011). Selecting the number of components in PCA using cross-validation approximations. *Computational Statistics and Data Analysis*. 56 (6), pp. 1869-1879.

# Percentage of inertia obtained under independence

$\Rightarrow$ Is there any structure in the data?

| | Number of variables | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Nb obs | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 5 | 96.5 | 93.1 | 90.2 | 87.6 | 85.5 | 83.4 | 81.9 | 80.7 | 79.4 | 78.1 | 77.4 | 76.6 | 75.5 |
| 6 | 93.3 | 88.6 | 84.8 | 81.5 | 79.1 | 76.9 | 75.1 | 73.2 | 72.2 | 70.8 | 69.8 | 68.7 | 68.0 |
| 7 | 90.5 | 84.9 | 80.9 | 77.4 | 74.4 | 72.0 | 70.1 | 68.3 | 67.0 | 65.3 | 64.3 | 63.2 | 62.2 |
| 8 | 88.1 | 82.3 | 77.2 | 73.8 | 70.7 | 68.2 | 66.1 | 64.0 | 62.8 | 61.2 | 60.0 | 59.0 | 58.0 |
| 9 | 86.1 | 79.5 | 74.8 | 70.7 | 67.4 | 65.1 | 62.9 | 61.1 | 59.4 | 57.9 | 56.5 | 55.4 | 54.3 |
| 10 | 84.5 | 77.5 | 72.3 | 68.2 | 65.0 | 62.4 | 60.1 | 58.3 | 56.5 | 55.1 | 53.7 | 52.5 | 51.5 |
| 11 | 82.8 | 75.7 | 70.3 | 66.3 | 62.9 | 60.1 | 58.0 | 56.0 | 54.4 | 52.7 | 51.3 | 50.1 | 49.2 |
| 12 | 81.5 | 74.0 | 68.6 | 64.4 | 61.2 | 58.3 | 55.8 | 54.0 | 52.4 | 50.9 | 49.3 | 48.2 | 47.2 |
| 13 | 80.0 | 72.5 | 67.2 | 62.9 | 59.4 | 56.7 | 54.4 | 52.2 | 50.5 | 48.9 | 47.7 | 46.6 | 45.4 |
| 14 | 79.0 | 71.5 | 65.7 | 61.5 | 58.1 | 55.1 | 52.8 | 50.8 | 49.0 | 47.5 | 46.2 | 45.0 | 44.0 |
| 15 | 78.1 | 70.3 | 64.6 | 60.3 | 57.0 | 53.9 | 51.5 | 49.4 | 47.8 | 46.1 | 44.9 | 43.6 | 42.5 |
| 16 | 77.3 | 69.4 | 63.5 | 59.2 | 55.6 | 52.9 | 50.3 | 48.3 | 46.6 | 45.2 | 43.6 | 42.4 | 41.4 |
| 17 | 76.5 | 68.4 | 62.6 | 58.2 | 54.7 | 51.8 | 49.3 | 47.1 | 45.5 | 44.0 | 42.6 | 41.4 | 40.3 |
| 18 | 75.5 | 67.6 | 61.8 | 57.1 | 53.7 | 50.8 | 48.4 | 46.3 | 44.6 | 43.0 | 41.6 | 40.4 | 39.3 |
| 19 | 75.1 | 67.0 | 60.9 | 56.5 | 52.8 | 49.9 | 47.4 | 45.5 | 43.7 | 42.1 | 40.7 | 39.6 | 38.4 |
| 20 | 74.1 | 66.1 | 60.1 | 55.6 | 52.1 | 49.1 | 46.6 | 44.7 | 42.9 | 41.3 | 39.8 | 38.7 | 37.5 |
| 25 | 72.0 | 63.3 | 57.1 | 52.5 | 48.9 | 46.0 | 43.4 | 41.4 | 39.6 | 38.1 | 36.7 | 35.5 | 34.5 |
| 30 | 69.8 | 61.1 | 55.1 | 50.3 | 46.7 | 43.6 | 41.1 | 39.1 | 37.3 | 35.7 | 34.4 | 33.2 | 32.1 |
| 35 | 68.5 | 59.6 | 53.3 | 48.6 | 44.9 | 41.9 | 39.5 | 37.4 | 35.6 | 34.0 | 32.7 | 31.6 | 30.4 |
| 40 | 67.5 | 58.3 | 52.0 | 47.3 | 43.4 | 40.5 | 38.0 | 36.0 | 34.1 | 32.7 | 31.3 | 30.1 | 29.1 |
| 45 | 66.4 | 57.1 | 50.8 | 46.1 | 42.4 | 39.3 | 36.9 | 34.8 | 33.1 | 31.5 | 30.2 | 29.0 | 27.9 |
| 50 | 65.6 | 56.3 | 49.9 | 45.2 | 41.4 | 38.4 | 35.9 | 33.9 | 32.1 | 30.5 | 29.2 | 28.1 | 27.0 |
| 100 | 60.9 | 51.4 | 44.9 | 40.0 | 36.3 | 33.3 | 31.0 | 28.9 | 27.2 | 25.8 | 24.5 | 23.3 | 22.3 |

# Percentage of inertia obtained under independence

| | Number of variables | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Nb ind | 17 | 18 | 19 | 20 | 25 | 30 | 35 | 40 | 50 | 75 | 100 | 150 | 200 |
| 5 | 74.9 | 74.2 | 73.5 | 72.8 | 70.7 | 68.8 | 67.4 | 66.4 | 64.7 | 62.0 | 60.5 | 58.5 | 57.4 |
| 6 | 67.0 | 66.3 | 65.6 | 64.9 | 62.3 | 60.4 | 58.9 | 57.6 | 55.8 | 52.9 | 51.0 | 49.0 | 47.8 |
| 7 | 61.3 | 60.7 | 59.7 | 59.1 | 56.4 | 54.3 | 52.6 | 51.4 | 49.5 | 46.4 | 44.6 | 42.4 | 41.2 |
| 8 | 57.0 | 56.2 | 55.4 | 54.5 | 51.8 | 49.7 | 47.8 | 46.7 | 44.6 | 41.6 | 39.8 | 37.6 | 36.4 |
| 9 | 53.6 | 52.5 | 51.8 | 51.2 | 48.1 | 45.9 | 44.4 | 42.9 | 41.0 | 38.0 | 36.1 | 34.0 | 32.7 |
| 10 | 50.6 | 49.8 | 49.0 | 48.3 | 45.2 | 42.9 | 41.4 | 40.1 | 38.0 | 35.0 | 33.2 | 31.0 | 29.8 |
| 11 | 48.1 | 47.2 | 46.5 | 45.8 | 42.8 | 40.6 | 39.0 | 37.7 | 35.6 | 32.6 | 30.8 | 28.7 | 27.5 |
| 12 | 46.2 | 45.2 | 44.4 | 43.8 | 40.7 | 38.5 | 36.9 | 35.5 | 33.5 | 30.5 | 28.8 | 26.7 | 25.5 |
| 13 | 44.4 | 43.4 | 42.8 | 41.9 | 39.0 | 36.8 | 35.1 | 33.9 | 31.8 | 28.8 | 27.1 | 25.0 | 23.9 |
| 14 | 42.9 | 42.0 | 41.3 | 40.4 | 37.4 | 35.2 | 33.6 | 32.3 | 30.4 | 27.4 | 25.7 | 23.6 | 22.4 |
| 15 | 41.6 | 40.7 | 39.8 | 39.1 | 36.2 | 34.0 | 32.4 | 31.1 | 29.0 | 26.0 | 24.3 | 22.4 | 21.2 |
| 16 | 40.4 | 39.5 | 38.7 | 37.9 | 35.0 | 32.8 | 31.1 | 29.8 | 27.9 | 24.9 | 23.2 | 21.2 | 20.1 |
| 17 | 39.4 | 38.5 | 37.6 | 36.9 | 33.8 | 31.7 | 30.1 | 28.8 | 26.8 | 23.9 | 22.2 | 20.3 | 19.2 |
| 18 | 38.3 | 37.4 | 36.7 | 35.8 | 32.9 | 30.7 | 29.1 | 27.8 | 25.9 | 22.9 | 21.3 | 19.4 | 18.3 |
| 19 | 37.4 | 36.5 | 35.8 | 34.9 | 32.0 | 29.9 | 28.3 | 27.0 | 25.1 | 22.2 | 20.5 | 18.6 | 17.5 |
| 20 | 36.7 | 35.8 | 34.9 | 34.2 | 31.3 | 29.1 | 27.5 | 26.2 | 24.3 | 21.4 | 19.8 | 18.0 | 16.9 |
| 25 | 33.5 | 32.5 | 31.8 | 31.1 | 28.1 | 26.0 | 24.5 | 23.3 | 21.4 | 18.6 | 17.0 | 15.2 | 14.2 |
| 30 | 31.2 | 30.3 | 29.5 | 28.8 | 26.0 | 23.9 | 22.3 | 21.1 | 19.3 | 16.6 | 15.1 | 13.4 | 12.5 |
| 35 | 29.5 | 28.6 | 27.9 | 27.1 | 24.3 | 22.2 | 20.7 | 19.6 | 17.8 | 15.2 | 13.7 | 12.1 | 11.1 |
| 40 | 28.1 | 27.3 | 26.5 | 25.8 | 23.0 | 21.0 | 19.5 | 18.4 | 16.6 | 14.1 | 12.7 | 11.1 | 10.2 |
| 45 | 27.0 | 26.1 | 25.4 | 24.7 | 21.9 | 20.0 | 18.5 | 17.4 | 15.7 | 13.2 | 11.8 | 10.3 | 9.4 |
| 50 | 26.1 | 25.3 | 24.6 | 23.8 | 21.1 | 19.1 | 17.7 | 16.6 | 14.9 | 12.5 | 11.1 | 9.6 | 8.7 |
| 100 | 21.5 | 20.7 | 19.9 | 19.3 | 16.7 | 14.9 | 13.6 | 12.5 | 11.0 | 8.9 | 7.7 | 6.4 | 5.7 |



**Table 2:** 95 % quantile inertia on the two first dimensions of 10000 PCA

# Quality of the representation: $\cos^2(\theta)$

- observations: $cos^2$ entre $O_i$ et $OF_i$

```
round(res.pca$ind$cos2,2)
                 Dim.1 Dim.2
S Michaud         0.62  0.07
S Renaudie        0.73  0.15
S Trotignon       0.78  0.07
```

- variable: $cos^2$ between a variable and its projection

```
round(res.pca$var$cos2,2)
                              Dim.1 Dim.2
Odor.Intensity.before.shaking  0.01  0.94
Odor.Intensity.after.shaking   0.01  0.89
Expression                     0.11  0.71
```

$\Rightarrow$ Only well projected elements can be interpreted

# Contribution

$\Rightarrow$ Contribution to the dimension (percentage of variability)

- observation: $\text{Ctr}_q(x_{i.}) = \frac{(1/I)F_{iq}^2}{\sum_{i=1}^{I}(1/I)F_{iq}^2}$

  $\Rightarrow$ observations with large coordinate contribute the most

```
round(res.pca$ind$contrib,2)
                 Dim.1 Dim.2
S Michaud        15.49  3.10
S Renaudie       15.56  5.56
S Trotignon      15.46  2.43
```

- variable: $\text{Ctr}_q(x_{k.}) = \frac{G_{kq}^2}{\sum_{k=K} G_{kq}^2} = \frac{r(x_{.k},v_q)^2}{r(x_{.k},v_q)^2}$

  $\Rightarrow$ variables with large correlation contribute the most

# Description of dimensions

Using continuous variables:

- correlation between variable and the principal components
- sort correlation coefficients and give significant ones (rought tests)

```
> dimdesc(res.pca)

            $Dim.1$quanti                                $Dim.2$quanti
                 corr p.value                                  corr p.value
O.candied.fruit  0.93 9.5e-05  Odor.Intensity.before.shaking 0.97 3.1e-06
Grade            0.93 1.2e-04  Odor.Intensity.after.shaking  0.95 3.6e-05
Surface.feeling  0.89 5.5e-04  Attack.intensity              0.85 1.7e-03
Typicity         0.86 1.4e-03  Expression                    0.84 2.2e-03
O.mushroom       0.84 2.3e-03  Aroma.persistency             0.75 1.3e-02
Visual.intensity 0.83 3.1e-03  Bitterness                    0.71 2.3e-02
   ...            ...     ...  Aroma.intensity               0.66 4.0e-02
O.plante        -0.87 1.0e-03
O.flower        -0.89 4.9e-04
O.passion       -0.90 4.5e-04
```

# Description of dimensions

Using categorical variables:

- One-way anova with the coordinates of the observations ($F_{.q}$) explained by the categorical variable
  - F-test by variable
  - for each category, a Student's $T$-test to compare the average of the category with the general mean

```
> dimdesc(res.pca)

Dim.1$quali
             R2      p.value
Label     0.874     7.30e-05

Dim.1$category
           Estimate      p.value
Vouvray       3.203     7.30e-05
Sauvignon    -3.203     7.30e-05
```

# Practice with R

1. Select active variables
2. Scale or not the variables
3. Perform PCA
4. Choose the number of dimensions to interpret
5. Simultaneously interpret the observations and variables graphs
6. Use interpretation tools

```
library(FactoMineR)
Expert <- read.table("http://factominer.free.fr/course/doc/data_PCA_ExpertWine.
   header = TRUE, sep = ";", row.names = 1)
res.pca <- PCA(Expert, scale = T, quanti.sup = 29:30, quali.sup = 1)
summary(res.pca)
barplot(res.pca$eig[,1], main = "Eigenvalues", names.arg = 1:nrow(res.pca$eig))
plot.PCA(res.pca, habillage = 1)
res.pca$ind$coord; res.pca$ind$cos2; res.pca$ind$contrib
plot.PCA(res.pca, axes = c(3, 4), habillage = 1)
dimdesc(res.pca)
plotellipses(res.pca, 1)
```

# References FactoMineR

*Exploratory Multivariate Analysis by Example using R*, Husson, Le, Pages (2017), Chapman & Hall

*Multiple Factor Analysis by Example using R*, Pages (2015), CRC Press

Package `FactoMineR`: `http://factominer.free.fr`

Youtube: playlist

MOOC

`https://husson.github.io/MOOC_AnaDo/index.html`

Playlist en Français.

FactoShiny.

FactoInvestigate

Données manquantes

**Previous topic: exploration with PCA**

- similarities between observations
- linear relationship between variables
- description of observations with variables
- small number of synthetic variables
- Vizualise/ **Dimension reduction**: represent data in a lower dimension space, keeping information as much as possible

**Now: Clustering. Hierarchical Clustering - K-means**

- Finding *homogeneous groups*, or clusters, in a data set

Both approaches share similarities: study links between objects - variability of the data. Different representation.
**Unsupervised learning** setting: no target variable
We aim at uncovering (learning) what is hidden in the data

# Motivations for clustering

- **Marketing:** find groups of customers with similar behavior who might be more receptive to a particular form of advertising, or more likely to purchase a particular product.
  Ex: L'OREAL: 100 000 women different countries - 300 questions
  • questionnaire: life style, skin and hair characteristics, care and consumer habits
  • Clinical assessments by a dermatologist: facial skin complexion, wrinkles, scalp dryness, greasiness
  • Hair assessments by a hair dresser: abundance, volume, breakage

# Motivations for clustering

- **Marketing:** find groups of customers with similar behavior who might be more receptive to a particular form of advertising, or more likely to purchase a particular product.
  Ex: L'OREAL: 100 000 women different countries - 300 questions
  • questionnaire: life style, skin and hair characteristics, care and consumer habits
  • Clinical assessments by a dermatologist: facial skin complexion, wrinkles, scalp dryness, greasiness
  • Hair assessments by a hair dresser: abundance, volume, breakage
- **Biology:** tissue samples for patients with breast cancer described by gene expression measurements. There may be different unknown types of cancer which we could discover
- **Internet:** document classification, clustering weblog data to discover groups of similar access patterns

# Clustering

**Data** $D_I = \{x_{1.}, \ldots, x_{I.}\}$ with $x_i \in \mathbb{R}^K$

**Goal** : retrieve groups = clusters = classes of individuals

2 individuals within a group must be as similar as possible

2 individuals of different groups must be as different as possible

Construct a map $f : \mathbb{R}^K \to \{1, \ldots, Q\}$ which affects a cluster number to each $x_{i.}$: $f : x_{i.} \mapsto q_i$

**No ground truth** for $q_i$... **Warning:** Choice of $Q$ is hard

# Clustering

**Data** $D_I = \{x_{1.}, \ldots, x_{I.}\}$ with $x_i \in \mathbb{R}^K$

**Goal** : retrieve groups = clusters = classes of individuals
2 individuals within a group must be as similar as possible
2 individuals of different groups must be as different as possible
Construct a map $f : \mathbb{R}^K \to \{1, \ldots, Q\}$ which affects a cluster number to each $x_{i.}$: $f : x_{i.} \mapsto q_i$
**No ground truth** for $q_i$... **Warning:** Choice of $Q$ is hard

Strengths: can be applied to objects with various features. Here: objects are statistical obs described by variables.

# Clustering

**Data** $D_I = \{x_{1.}, \ldots, x_{I.}\}$ with $x_i \in \mathbb{R}^K$

**Goal** : retrieve groups = clusters = classes of individuals
2 individuals within a group must be as similar as possible
2 individuals of different groups must be as different as possible
Construct a map $f : \mathbb{R}^K \to \{1, \ldots, Q\}$ which affects a cluster number to each $x_{i.}$: $f : x_{i.} \mapsto q_i$
**No ground truth** for $q_i$... **Warning:** Choice of $Q$ is hard

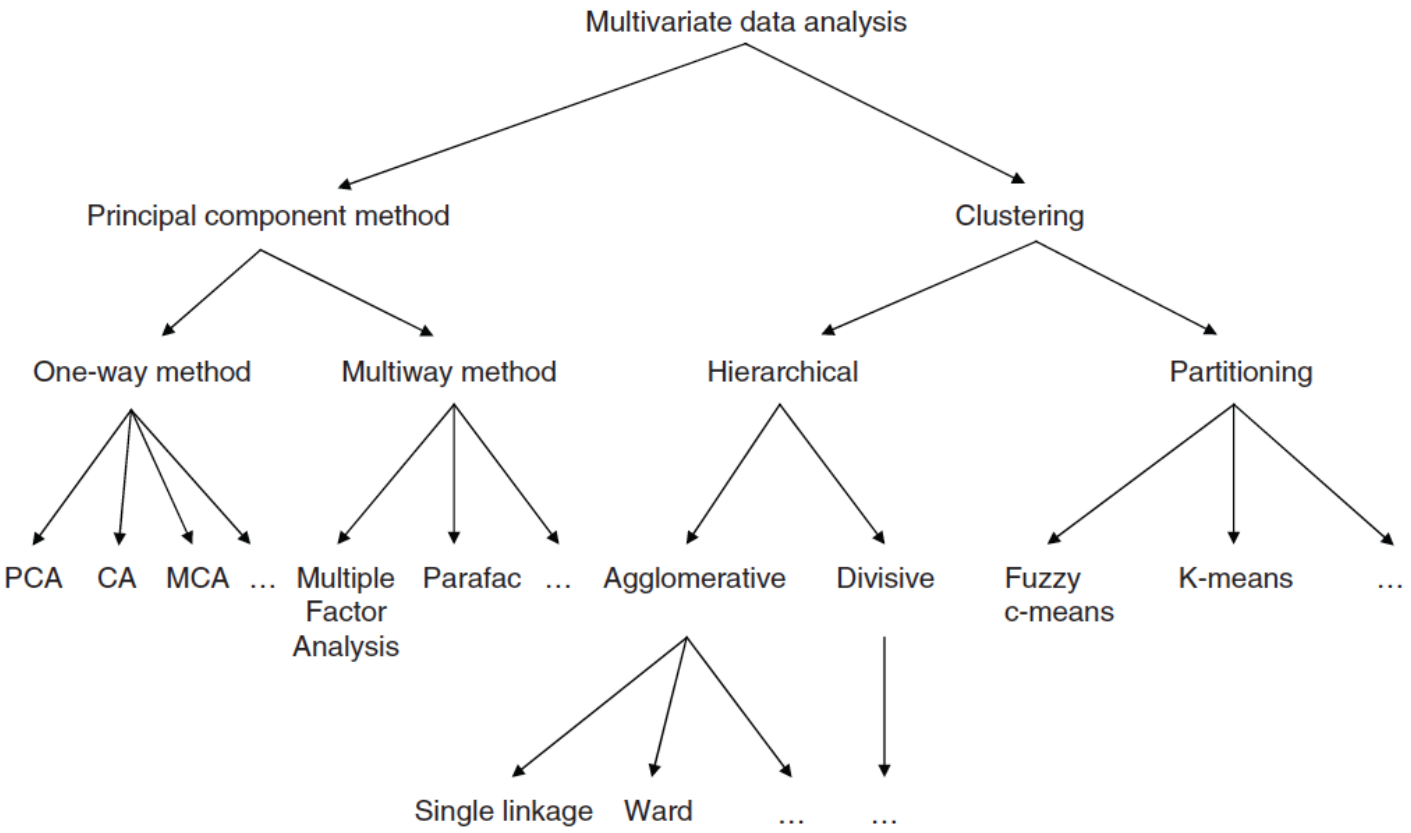
Strengths: can be applied to objects with various features. Here: objects are statistical obs described by variables.

There are many different types of clustering methods. We will concentrate on two of the most commonly used approaches

- Hierarchical Clustering - Partition-based
- Model-based (Mixture models - EM algo)

# Principles of hierarchical clustering

# Hierarchical example: unsupervised methods



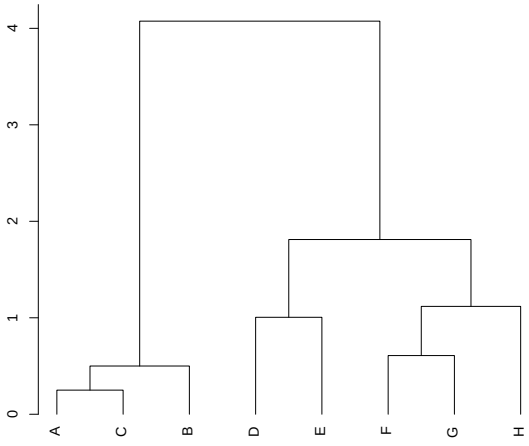
# What data? What goals?

Data tables: rows of individuals, columns of quantitative variables

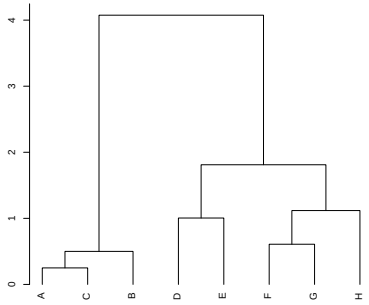
| | Janv | Févr | Mars | Avri | Mai | Juin | juil | Août | Sept | Octo | Nove | Déce | Lati | Long |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bordeaux | 5.6 | 6.6 | 10.3 | 12.8 | 15.8 | 19.3 | 20.9 | 21 | 18.6 | 13.8 | 9.1 | 6.2 | 44.5 | -0.34 |
| Brest | 6.1 | 5.8 | 7.8 | 9.2 | 11.6 | 14.4 | 15.6 | 16 | 14.7 | 12 | 9 | 7 | 48.24 | -4.29 |
| Clermont | 2.6 | 3.7 | 7.5 | 10.3 | 13.8 | 17.3 | 19.4 | 19.1 | 16.2 | 11.2 | 6.6 | 3.6 | 45.47 | 3.05 |
| Grenoble | 1.5 | 3.2 | 7.7 | 10.6 | 14.5 | 17.8 | 20.1 | 19.5 | 16.7 | 11.4 | 6.5 | 2.3 | 45.1 | 5.43 |
| Lille | 2.4 | 2.9 | 6 | 8.9 | 12.4 | 15.3 | 17.1 | 17.1 | 14.7 | 10.4 | 6.1 | 3.5 | 50.38 | 3.04 |
| Lyon | 2.1 | 3.3 | 7.7 | 10.9 | 14.9 | 18.5 | 20.7 | 20.1 | 16.9 | 11.4 | 6.7 | 3.1 | 45.45 | 4.51 |
| Marseille | 5.5 | 6.6 | 10 | 13 | 16.8 | 20.8 | 23.3 | 22.8 | 19.9 | 15 | 10.2 | 6.9 | 43.18 | 5.24 |
| Montpellier | 5.6 | 6.7 | 9.9 | 12.8 | 16.2 | 20.1 | 22.7 | 22.3 | 19.3 | 14.6 | 10 | 6.5 | 43.36 | 3.53 |
| Nantes | 5 | 5.3 | 8.4 | 10.8 | 13.9 | 17.2 | 18.8 | 18.6 | 16.4 | 12.2 | 8.2 | 5.5 | 47.13 | -1.33 |
| Nice | 7.5 | 8.5 | 10.8 | 13.3 | 16.7 | 20.1 | 22.7 | 22.5 | 20.3 | 16 | 11.5 | 8.2 | 43.42 | 7.15 |
| Paris | 3.4 | 4.1 | 7.6 | 10.7 | 14.3 | 17.5 | 19.1 | 18.7 | 16 | 11.4 | 7.1 | 4.3 | 48.52 | 2.2 |
| Rennes | 4.8 | 5.3 | 7.9 | 10.1 | 13.1 | 16.2 | 17.9 | 17.8 | 15.7 | 11.6 | 7.8 | 5.4 | 48.05 | -1.41 |
| Strasbourg | 0.4 | 1.5 | 5.6 | 9.8 | 14 | 17.2 | 19 | 18.3 | 15.1 | 9.5 | 4.9 | 1.3 | 48.35 | 7.45 |
| Toulouse | 4.7 | 5.6 | 9.2 | 11.6 | 14.9 | 18.7 | 20.9 | 20.9 | 18.3 | 13.3 | 8.6 | 5.5 | 43.36 | 1.26 |
| Vichy | 2.4 | 3.4 | 7.1 | 9.9 | 13.6 | 17.1 | 19.3 | 18.8 | 16 | 11 | 6.6 | 3.4 | 46.08 | 3.26 |

Goals: build a tree structure called a Dendogram that

- shows hierarchical links between individuals or groups of individuals
- detects a “natural” number of classes in the population



# Indexed hierarchy




**Simple interpretation**: Each leaf represents one of the observations. As we move up the tree, some leaves fuse into branches. These correspond to observations that are similar to each other. As we move higher up the tree, branches themselves fuse, either with leaves or other branches.

Height of fusing/merging (on vertical axis) indicates how similar the points are: 2 objects are similar if, to move from one to the other, it is not necessary to go too far back in the tree. (A & C) more similar than (F & G). H more similar to (D & E) than to (A & B).

**Rk**: the positions of the two fused branches could be swapped without affecting the meaning of the tree.

# Hierarchical clustering: principles

Starting from the leaves and combining clusters up to the trunk

- Start with $I$ initial clusters, each one being one observation. Agglomerate closest observations: smallest $d(i, i')$
- Sequentially merge the two closest clusters into one new cluster.
- Stop when you have only one cluster.

**Rk**: AHC first groups most similar objects and then groups of object. Divisive clustering

# Hierarchical clustering: principles

Starting from the leaves and combining clusters up to the trunk

- Start with $I$ initial clusters, each one being one observation. Agglomerate closest observations: smallest $d(i, i')$
- Sequentially merge the two closest clusters into one new cluster.
- Stop when you have only one cluster.

$\rightarrow$ How do we know which clusters are the closest from each other? We need to

- measure similarity of individuals (Euclidian to start)
- measure similarity between groups of individuals (for instance distance between $\{5, 8\}$ and 3?): linkage

**Rk**: AHC first groups most similar objects and then groups of object. Divisive clustering

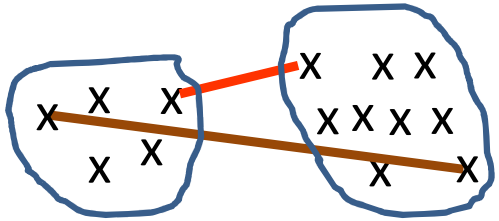
# Linkage criterion between sets $C_q$ and $C'_q$

Similarity between groups of individuals:

- Single linkage/Nearest neighbor (smallest distance):

$$d(C_q, C'_q) = \min\{d(x,z), x \in C_q, z \in C'_q\}$$

- Complete linkage/Furthest neighbor (largest distance):



$$d(C_q C'_q) = \max\{d(x,z), x \in C_q, z \in C'_q\}$$

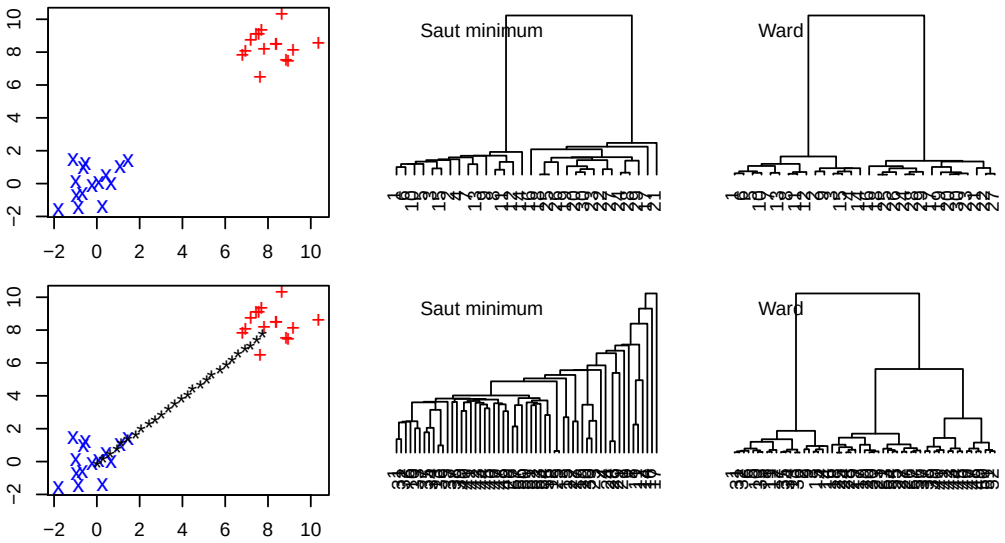
- Average linkage (average distance):

$$d(C_q, C'_q) = \frac{1}{|C_q||C'_q|} \sum_{x \in C_q, z \in C'_q} d(x,z)$$

Complete linkage preferred over single linkage: tends to yield more balanced dendrograms. No approach to choose the linkage from data

# Impact of the different approaches

Single linkage: chain effects




Group together classes with similar centers of gravity

Direct use for clustering

⇒ Small decisions with big consequences? (Scaling before)

⇒ Depends on the context (Read ex Element of Stat book)

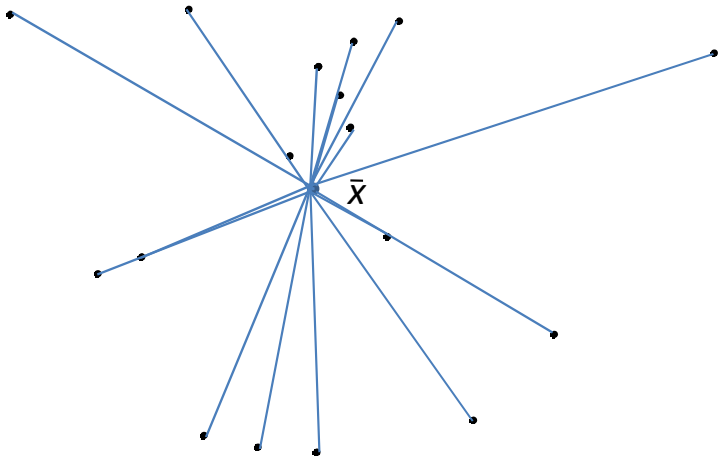
Try several different choices, and look for the one with the most useful or interpretable solution.

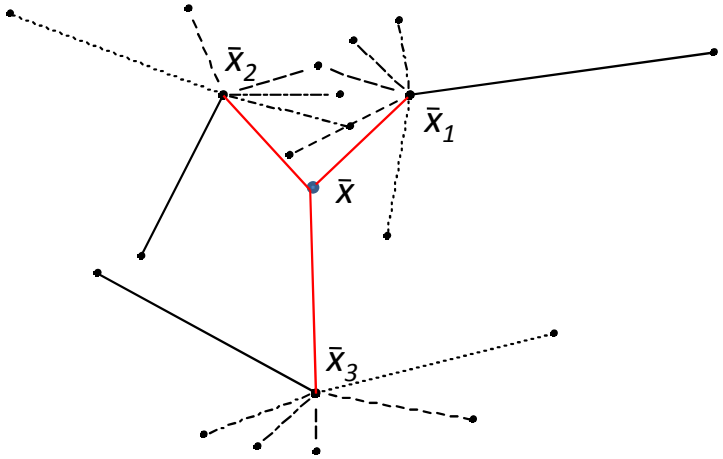
⇒ There is no single right answer

# Ward's method: Euclidian distances + Ward - Based on Inertia

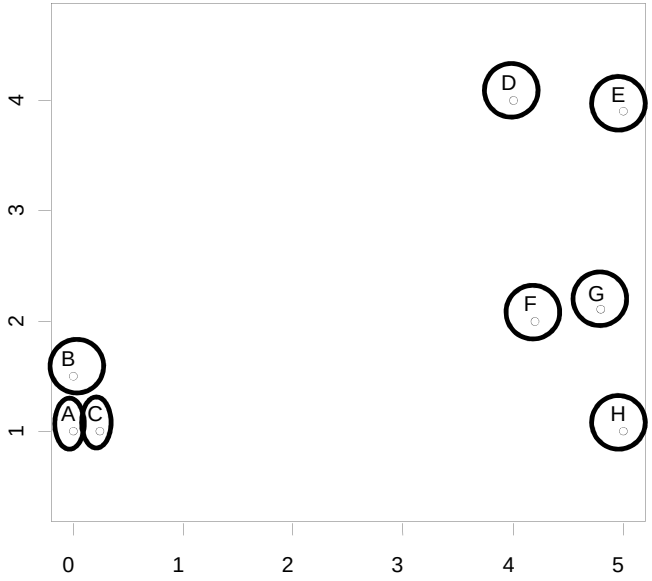
**Huygens theorem**: $\bar{x}_k$ the mean of variable $k$ $\bar{x}_k = \frac{1}{I}\sum_{i=1}^{I} x_{ik}$ , $\bar{x}_{qk}$ the mean of variable $k$ for the observations in cluster $q$
$\bar{x}_{qk} = \frac{1}{|C_q|}\sum_{i\in C_q} x_{ik} = \frac{1}{I_q}\sum_{i\in C_q} x_{ik}$

$$\underbrace{\frac{1}{I}\sum_{k=1}^{K}\sum_{q=1}^{Q}\sum_{i\in C_q}(x_{ik}-\bar{x}_k)^2}_{\text{total inertia}} = \underbrace{\frac{1}{I}\sum_{k=1}^{K}\sum_{q=1}^{Q}\sum_{i\in C_q}(x_{ik}-\bar{x}_{qk})^2}_{\text{within-class inertia}} + \underbrace{\frac{1}{I}\sum_{k=1}^{K}\sum_{q=1}^{Q}\sum_{i\in C_q}(\bar{x}_{qk}-\bar{x}_k)^2}_{\text{between-class inertia}}$$

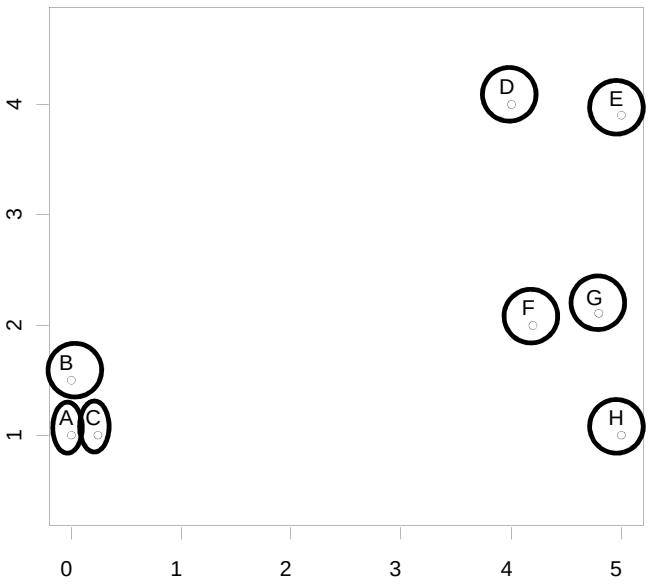


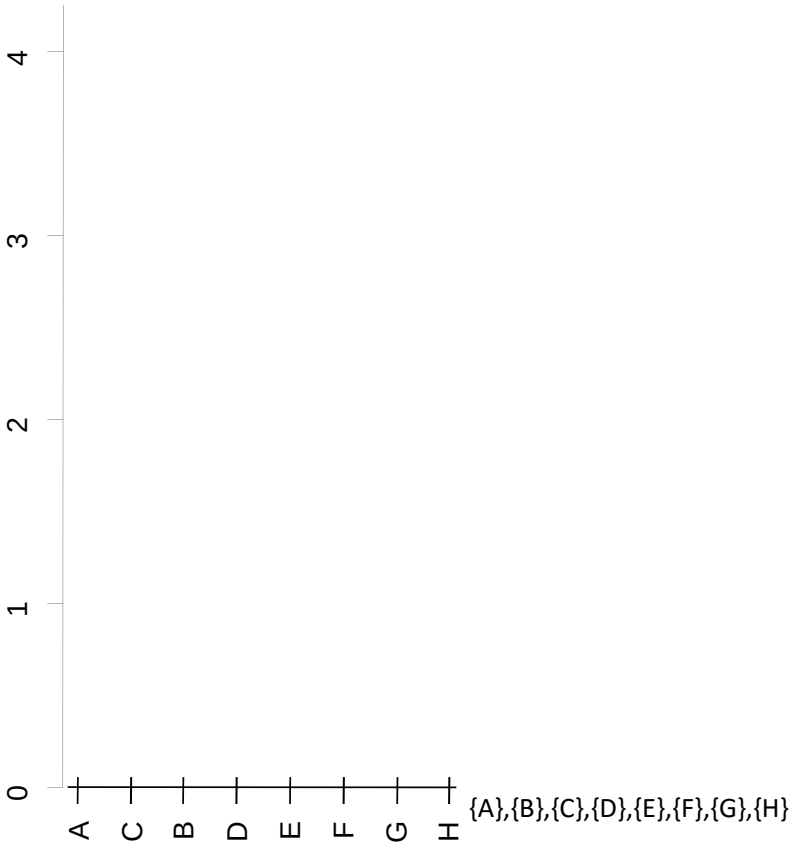



# Algorithm



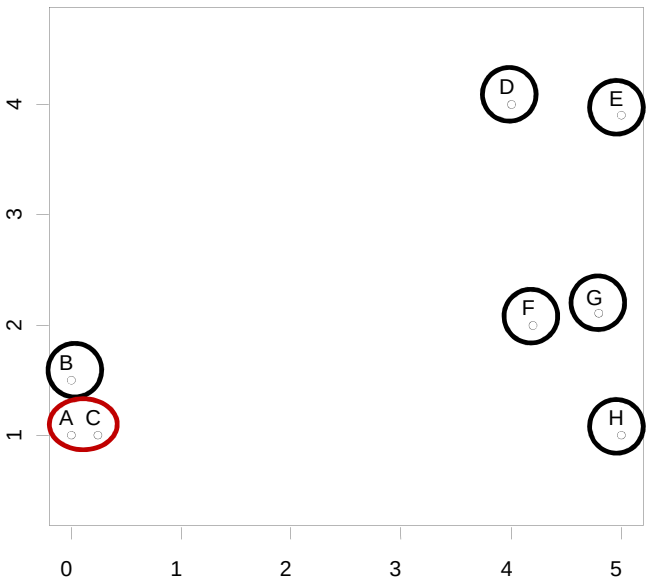
# Algorithm

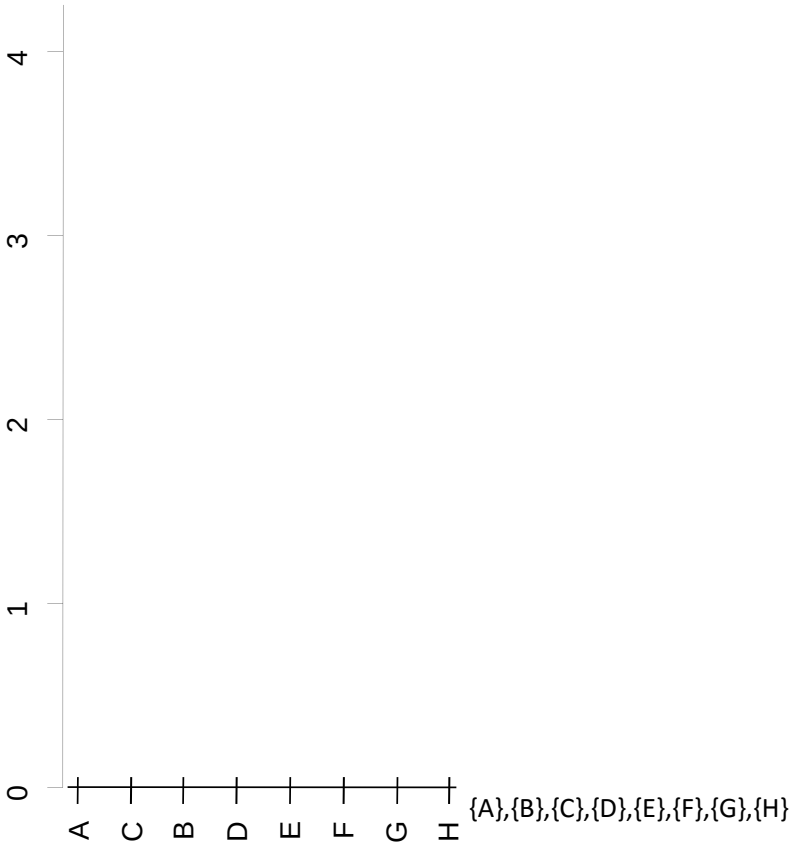



| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

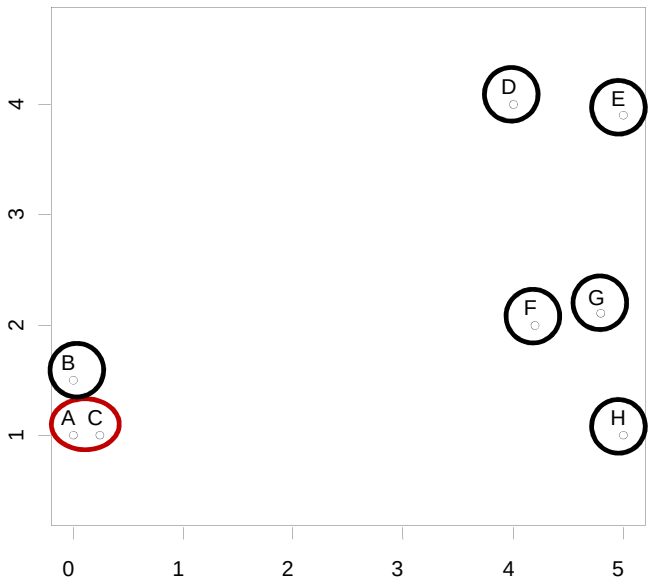
# Algorithm

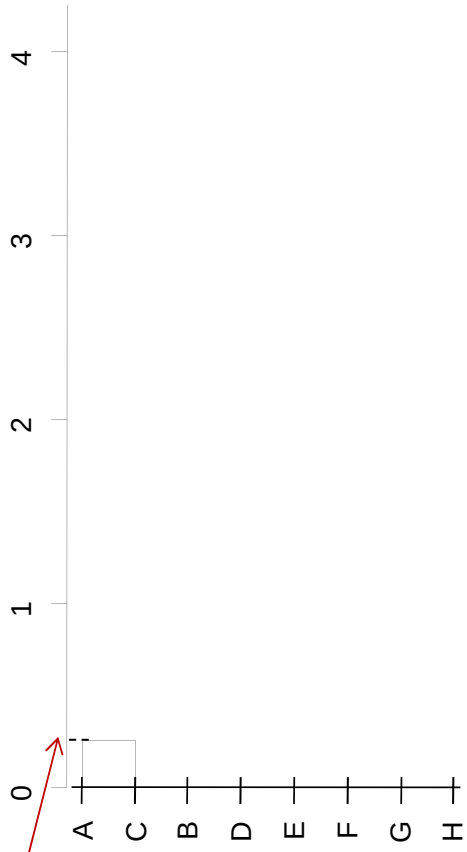



| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

# Algorithm



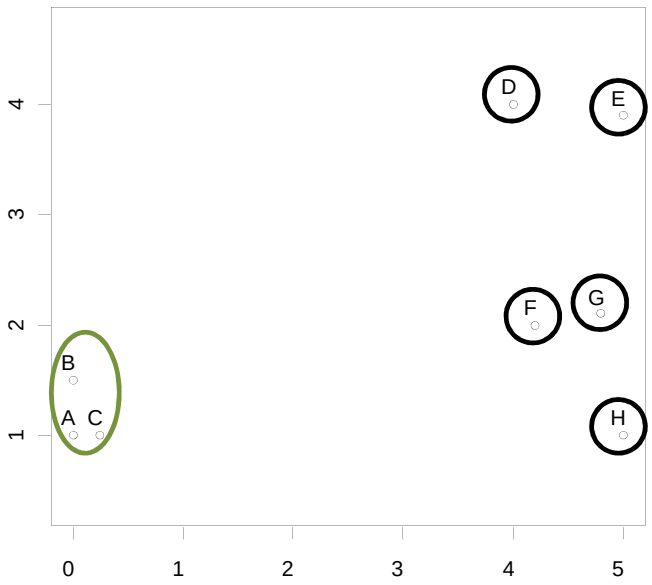

{AC},{B},{D},{E},{F},{G},{H}

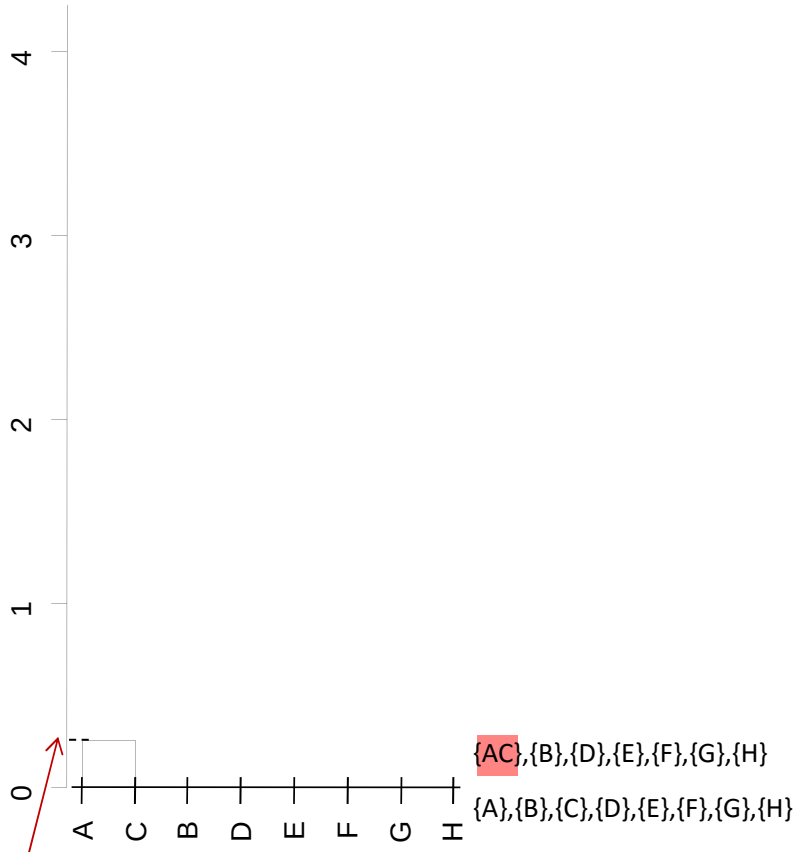
{A},{B},{C},{D},{E},{F},{G},{H}

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

# Algorithm



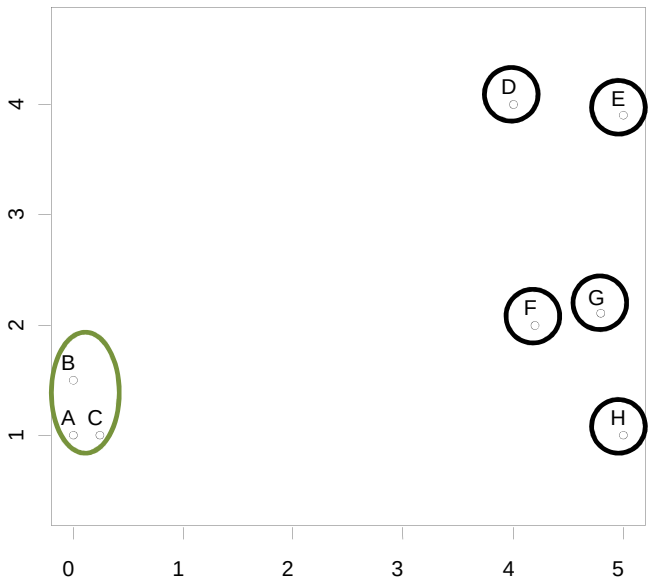



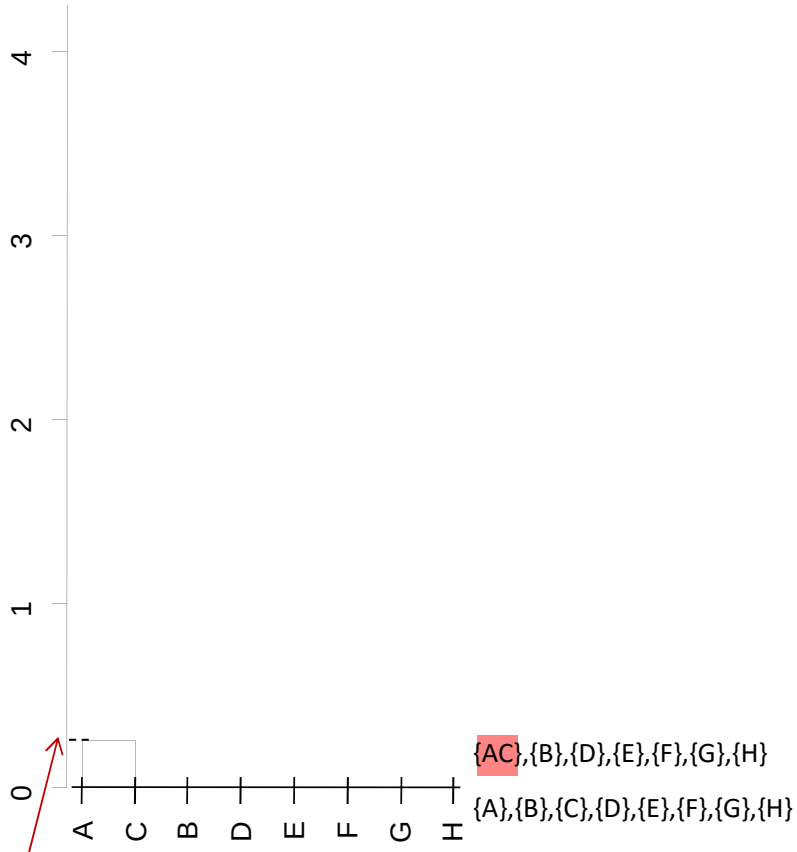

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

# Algorithm



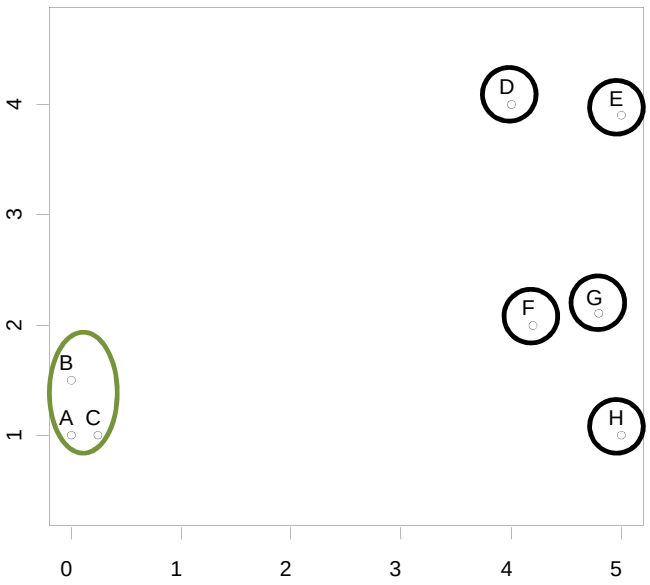

2nd grouping

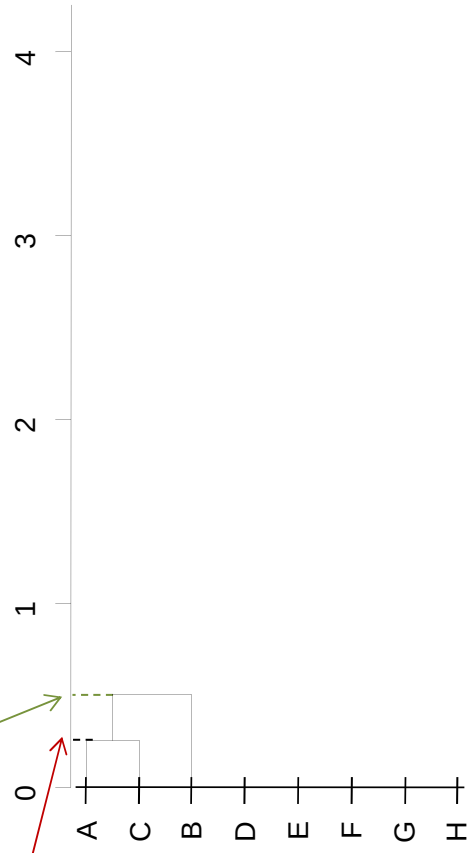
| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

# Algorithm




{ABC},{D},{E},{F},{G},{H}

{AC},{B},{D},{E},{F},{G},{H}

{A},{B},{C},{D},{E},{F},{G},{H}

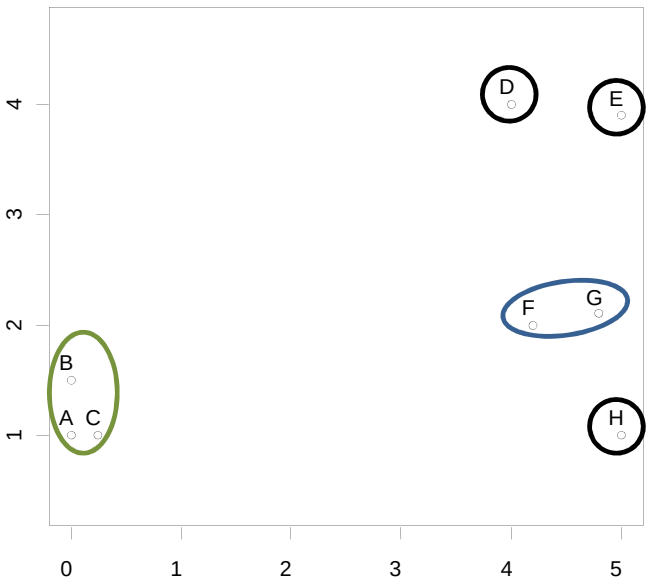
2nd grouping

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

# Algorithm



{ABC},{D},{E},{F},{G},{H}

{AC},{B},{D},{E},{F},{G},{H}

{A},{B},{C},{D},{E},{F},{G},{H}

3rd grouping

| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

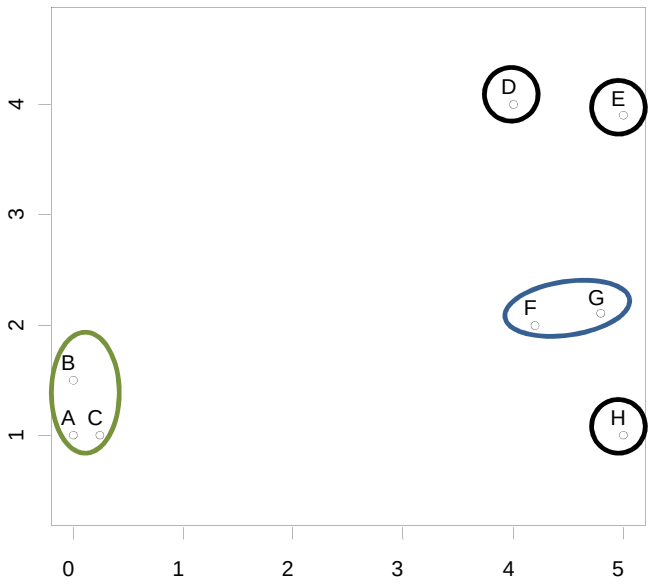
2nd grouping

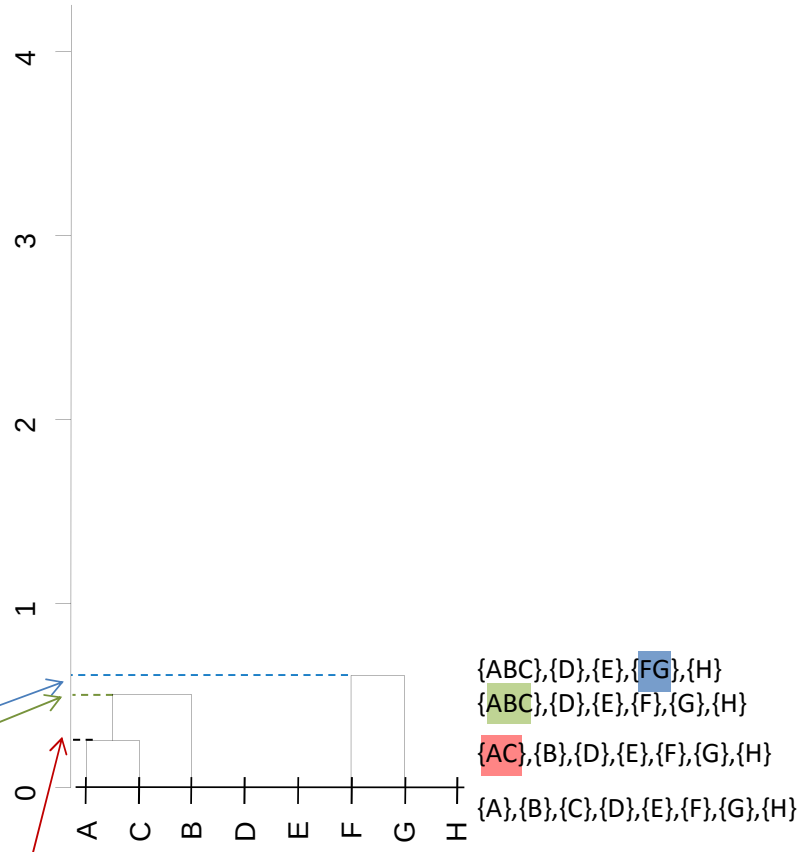
| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

# Algorithm




3rd grouping

| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

2nd grouping

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

# Algorithm

4th grouping

| | ABC | D | E | FG |
|---|---|---|---|---|
| D | 4.72 | | | |
| E | 5.55 | 1.00 | | |
| FG | 4.07 | 2.01 | 1.81 | |
| H | 4.75 | 3.16 | 2.90 | 1.12 |

3rd grouping

| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

2nd grouping

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

{ABC},{D},{E},{FG},{H}

{ABC},{D},{E},{F},{G},{H}

{AC},{B},{D},{E},{F},{G},{H}

{A},{B},{C},{D},{E},{F},{G},{H}

1st grouping

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

# Algorithm

4th grouping

| | ABC | D | E | FG |
|---|---|---|---|---|
| D | 4.72 | | | |
| E | 5.55 | 1.00 | | |
| FG | 4.07 | 2.01 | 1.81 | |
| H | 4.75 | 3.16 | 2.90 | 1.12 |

3rd grouping

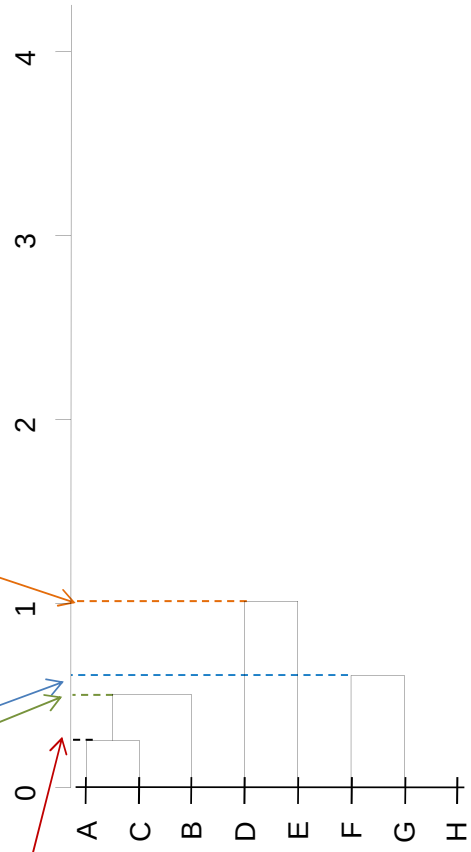
| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

2nd grouping

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |




{ABC},{DE},{FG},{H}

{ABC},{D},{E},{FG},{H}

{ABC},{D},{E},{F},{G},{H}

{AC},{B},{D},{E},{F},{G},{H}

{A},{B},{C},{D},{E},{F},{G},{H}

# Algorithm

5th grouping

| | ABC | DE | FG |
|---|---|---|---|
| DE | 4.72 | | |
| FG | 4.23 | 1.81 | |
| H | 4.07 | 2.90 | 1.12 |

4th grouping

| | ABC | D | E | FG |
|---|---|---|---|---|
| D | 4.72 | | | |
| E | 5.55 | 1.00 | | |
| FG | 4.07 | 2.01 | 1.81 | |
| H | 4.75 | 3.16 | 2.90 | 1.12 |

3rd grouping

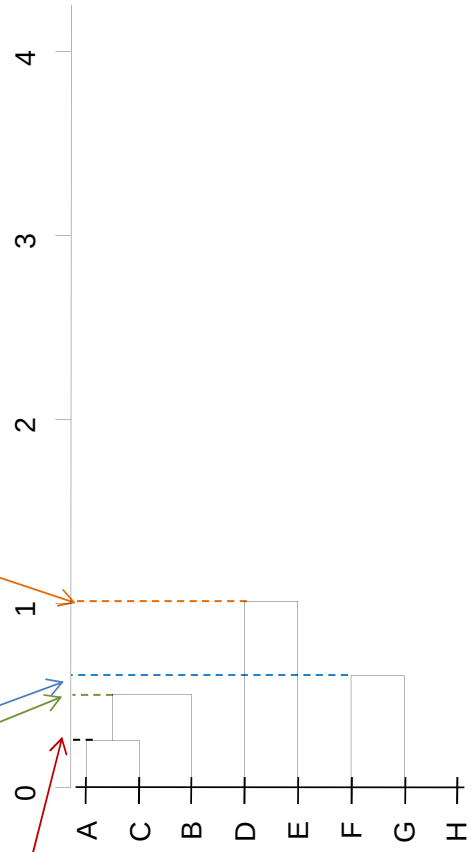
| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

2nd grouping

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |




{ABC},{DE},{FG},{H}

{ABC},{D},{E},{FG},{H}

{ABC},{D},{E},{F},{G},{H}

{AC},{B},{D},{E},{F},{G},{H}

{A},{B},{C},{D},{E},{F},{G},{H}

# Algorithm

5th grouping

4th grouping

3rd grouping

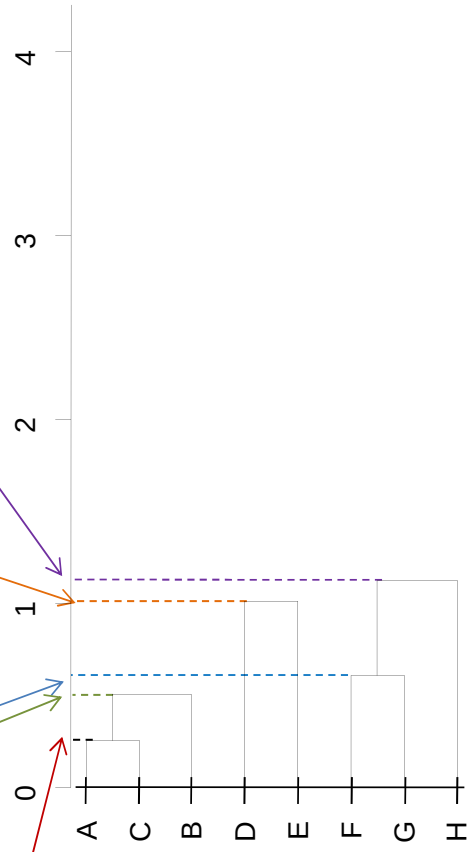
2nd grouping

| | ABC | DE | FG |
|---|---|---|---|
| DE | 4.72 | | |
| FG | 4.23 | 1.81 | |
| H | 4.07 | 2.90 | 1.12 |

| | ABC | D | E | FG |
|---|---|---|---|---|
| D | 4.72 | | | |
| E | 5.55 | 1.00 | | |
| FG | 4.07 | 2.01 | 1.81 | |
| H | 4.75 | 3.16 | 2.90 | 1.12 |

| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |




{ABC},{DE},{FGH}
{ABC},{DE},{FG},{H}
{ABC},{D},{E},{FG},{H}
{ABC},{D},{E},{F},{G},{H}
{AC},{B},{D},{E},{F},{G},{H}
{A},{B},{C},{D},{E},{F},{G},{H}

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

# Algorithm

6th grouping

5th grouping

4th grouping

3rd grouping

2nd grouping

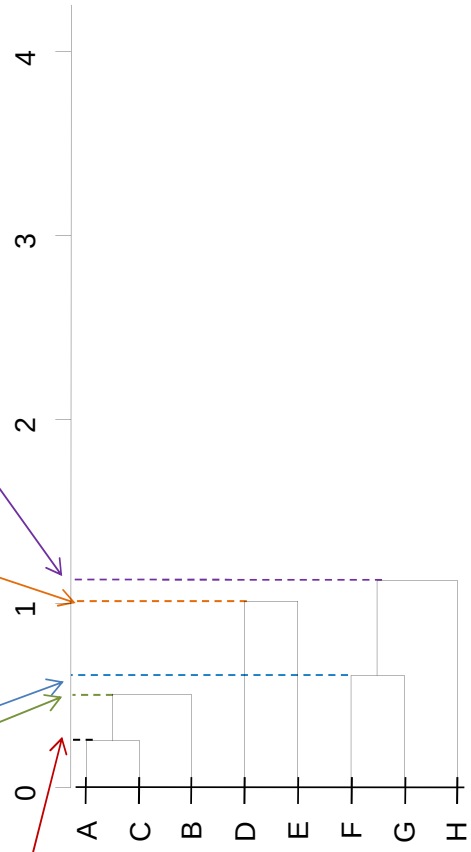
| | ABC | DE |
|---|---|---|
| DE | 4.72 | |
| FGH | 4.07 | 1.81 |

| | ABC | DE | FG |
|---|---|---|---|
| DE | 4.72 | | |
| FG | 4.23 | 1.81 | |
| H | 4.07 | 2.90 | 1.12 |

| | ABC | D | E | FG |
|---|---|---|---|---|
| D | 4.72 | | | |
| E | 5.55 | 1.00 | | |
| FG | 4.07 | 2.01 | 1.81 | |
| H | 4.75 | 3.16 | 2.90 | 1.12 |

| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |




{ABC},{DE},{FGH}
{ABC},{DE},{FG},{H}
{ABC},{D},{E},{FG},{H}
{ABC},{D},{E},{F},{G},{H}
{AC},{B},{D},{E},{F},{G},{H}
{A},{B},{C},{D},{E},{F},{G},{H}

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping

# Algorithm

6th grouping

5th grouping

4th grouping

3rd grouping

2nd grouping

| | ABC | DE |
|---|---|---|
| DE | 4.72 | |
| FGH | 4.07 | 1.81 |

| | ABC | DE | FG |
|---|---|---|---|
| DE | 4.72 | | |
| FG | 4.23 | 1.81 | |
| H | 4.07 | 2.90 | 1.12 |

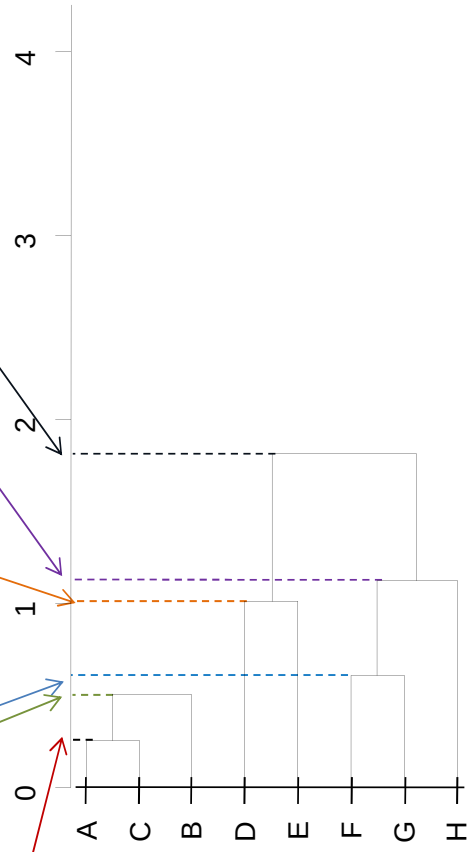
| | ABC | D | E | FG |
|---|---|---|---|---|
| D | 4.72 | | | |
| E | 5.55 | 1.00 | | |
| FG | 4.07 | 2.01 | 1.81 | |
| H | 4.75 | 3.16 | 2.90 | 1.12 |

| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping




{ABC},{DEFGH}

{ABC},{DE},{FGH}

{ABC},{DE},{FG},{H}

{ABC},{D},{E},{FG},{H}

{ABC},{D},{E},{F},{G},{H}

{AC},{B},{D},{E},{F},{G},{H}

{A},{B},{C},{D},{E},{F},{G},{H}

# Algorithm

7th grouping

6th grouping

5th grouping

4th grouping

3rd grouping

2nd grouping

| | ABC |
|---|---|
| DEFGH | 4.07 |

| | ABC | DE |
|---|---|---|
| DE | 4.72 | |
| FGH | 4.07 | 1.81 |

| | ABC | DE | FG |
|---|---|---|---|
| DE | 4.72 | | |
| FG | 4.23 | 1.81 | |
| H | 4.07 | 2.90 | 1.12 |

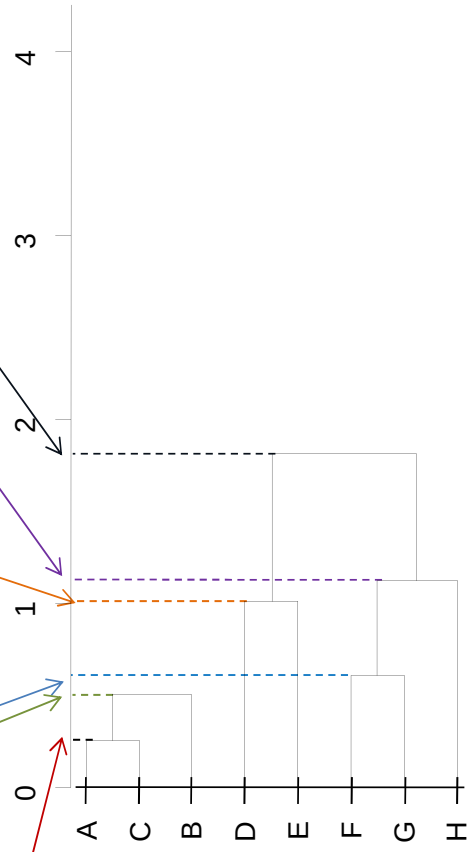
| | ABC | D | E | FG |
|---|---|---|---|---|
| D | 4.72 | | | |
| E | 5.55 | 1.00 | | |
| FG | 4.07 | 2.01 | 1.81 | |
| H | 4.75 | 3.16 | 2.90 | 1.12 |

| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping




{ABC},{DEFGH}

{ABC},{DE},{FGH}

{ABC},{DE},{FG},{H}

{ABC},{D},{E},{FG},{H}

{ABC},{D},{E},{F},{G},{H}

{AC},{B},{D},{E},{F},{G},{H}

{A},{B},{C},{D},{E},{F},{G},{H}

# Algorithm

7th grouping

6th grouping

5th grouping

4th grouping

3rd grouping

2nd grouping

| | ABC |
|---|---|
| DEFGH | 4.07 |

| | ABC | DE |
|---|---|---|
| DE | 4.72 | |
| FGH | 4.07 | 1.81 |

| | ABC | DE | FG |
|---|---|---|---|
| DE | 4.72 | | |
| FG | 4.23 | 1.81 | |
| H | 4.07 | 2.90 | 1.12 |

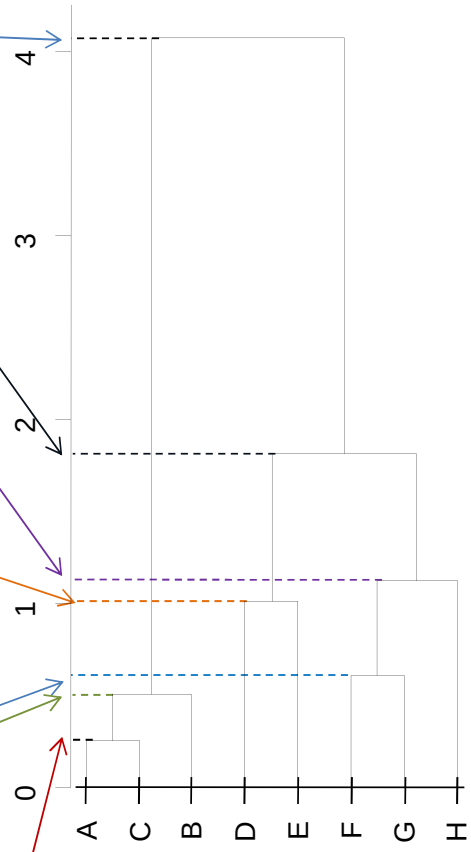
| | ABC | D | E | FG |
|---|---|---|---|---|
| D | 4.72 | | | |
| E | 5.55 | 1.00 | | |
| FG | 4.07 | 2.01 | 1.81 | |
| H | 4.75 | 3.16 | 2.90 | 1.12 |

| | ABC | D | E | F | G |
|---|---|---|---|---|---|
| D | 4.72 | | | | |
| E | 5.55 | 1.00 | | | |
| F | 4.07 | 2.01 | 2.06 | | |
| G | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

| | AC | B | D | E | F | G |
|---|---|---|---|---|---|---|
| B | 0.50 | | | | | |
| D | 4.80 | 4.72 | | | | |
| E | 5.57 | 5.55 | 1.00 | | | |
| F | 4.07 | 4.23 | 2.01 | 2.06 | | |
| G | 4.68 | 4.84 | 2.06 | 1.81 | 0.61 | |
| H | 4.75 | 5.02 | 3.16 | 2.90 | 1.28 | 1.12 |

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| B | 0.50 | | | | | | |
| C | 0.25 | 0.56 | | | | | |
| D | 5.00 | 4.72 | 4.80 | | | | |
| E | 5.78 | 5.55 | 5.57 | 1.00 | | | |
| F | 4.32 | 4.23 | 4.07 | 2.01 | 2.06 | | |
| G | 4.92 | 4.84 | 4.68 | 2.06 | 1.81 | 0.61 | |
| H | 5.00 | 5.02 | 4.75 | 3.16 | 2.90 | 1.28 | 1.12 |

1st grouping




{ABCDEFGH}

{ABC},{DEFGH}

{ABC},{DE},{FGH}

{ABC},{DE},{FG},{H}

{ABC},{D},{E},{FG},{H}

{ABC},{D},{E},{F},{G},{H}

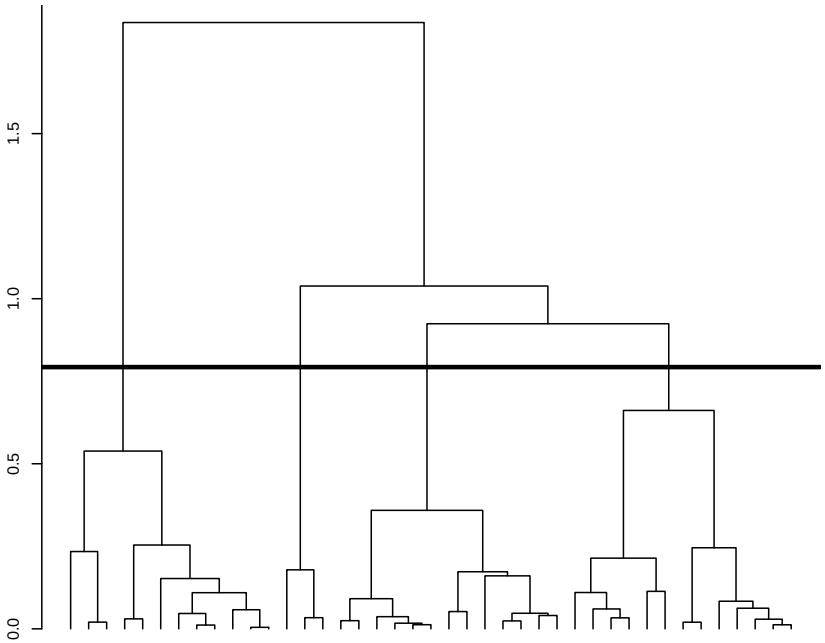
{AC},{B},{D},{E},{F},{G},{H}

{A},{B},{C},{D},{E},{F},{G},{H}

# Trees and partitions

Trees always end up . . . cut through!

Choosing a height to cut at gives a partition
Different cut, different partitions



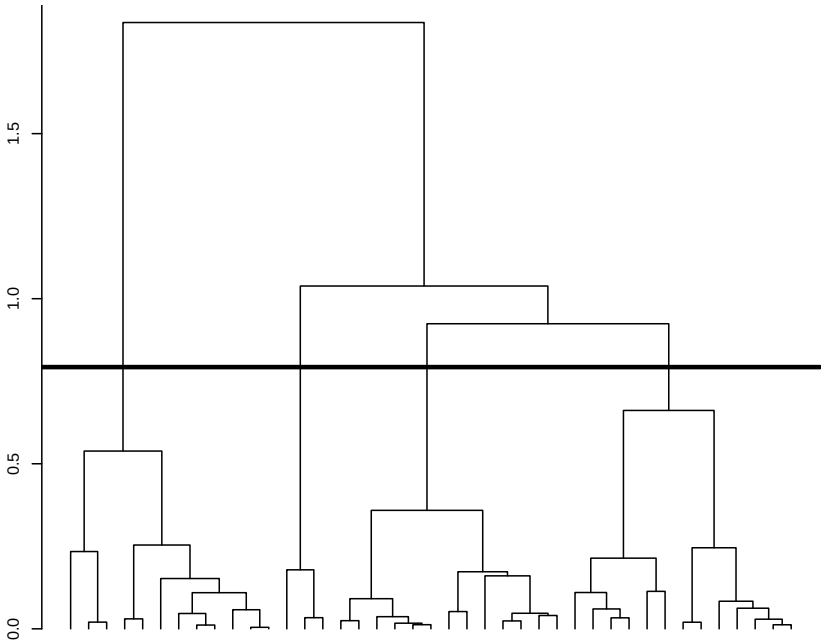

Hierarchical tree: sequence of **nested partitions** from most precise (each obs is a class) to most general (only 1 class)

**Advantage**: don't know in advance how many clusters - view at once the clusterings obtained from 1 to $I$ clusters.

# Trees and partitions

Trees always end up . . . cut through!

Choosing a height to cut at gives a partition
Different cut, different partitions




Hierarchical tree: sequence of **nested partitions** from most precise (each obs is a class) to most general (only 1 class)

**Advantage**: don't know in advance how many clusters - view at once the clusterings obtained from 1 to $I$ clusters.



Rk: the partition is interesting but not optimal

# Partition quality

When is a partition a good one?

# Partition quality

When is a partition a good one?

- If individuals placed in the same class are close to each other
- If individuals in different classes are far from each other

# Partition quality

When is a partition a good one?

- If individuals placed in the same class are close to each other
- If individuals in different classes are far from each other

Mathematically speaking?

- small within-class variability
- large between-class variability

# Partition quality

When is a partition a good one?

- If individuals placed in the same class are close to each other
- If individuals in different classes are far from each other

Mathematically speaking?

- small within-class variability
- large between-class variability

$$\underbrace{\frac{1}{I}\sum_{k=1}^{K}\sum_{q=1}^{Q}\sum_{i\in C_q}(x_{ik}-\bar{x}_k)^2}_{\text{total inertia}} = \underbrace{\frac{1}{I}\sum_{k=1}^{K}\sum_{q=1}^{Q}\sum_{i\in C_q}(x_{iqk}-\bar{x}_{qk})^2}_{\text{within-class inertia}} + \underbrace{\frac{1}{I}\sum_{k=1}^{K}\sum_{q=1}^{Q}\sum_{i\in C_q}(\bar{x}_{qk}-\bar{x}_k)^2}_{\text{between-class inertia}}$$

# Partition quality

Partition quality is measured by:

$$0 \leq \frac{\text{between-class inertia}}{\text{total inertia}} \leq 1$$

# Partition quality

Partition quality is measured by:

$$0 \leq \frac{\text{between-class inertia}}{\text{total inertia}} \leq 1$$

$$\frac{\text{between inertia}}{\text{total inertia}} = 0 \Longrightarrow \forall k, \forall q, \bar{x}_{qk} = \bar{x}_k$$

by variable, classes have the same means

Doesn't allow us to classify

# Partition quality

Partition quality is measured by:

$$0 \leq \frac{\text{between-class inertia}}{\text{total inertia}} \leq 1$$

$$\frac{\text{between inertia}}{\text{total inertia}} = 0 \Longrightarrow \forall k, \forall q, \bar{x}_{qk} = \bar{x}_k$$

by variable, classes have the same means

Doesn't allow us to classify

$$\frac{\text{between inertia}}{\text{total inertia}} = 1 \Longrightarrow \forall k, \forall q, \forall i, x_{iqk} = \bar{x}_{qk}$$

individuals in the same class are identical

Ideal for classifying

# Partition quality

Partition quality is measured by:

$$0 \leq \frac{\text{between-class inertia}}{\text{total inertia}} \leq 1$$

$$\frac{\text{between inertia}}{\text{total inertia}} = 0 \Longrightarrow \forall k, \forall q, \bar{x}_{qk} = \bar{x}_k$$

by variable, classes have the same means

Doesn't allow us to classify

$$\frac{\text{between inertia}}{\text{total inertia}} = 1 \Longrightarrow \forall k, \forall q, \forall i, x_{iqk} = \bar{x}_{qk}$$

individuals in the same class are identical

Ideal for classifying

It corresponds with 1 variable to the $\eta^2$ (or $R^2$ in analysis of variance) $\rightarrow$ percentage of variability explained by the partition (categorical variable).

# Partition quality

Partition quality is measured by:

$$0 \leq \frac{\text{between-class inertia}}{\text{total inertia}} \leq 1$$

$$\frac{\text{between inertia}}{\text{total inertia}} = 0 \Longrightarrow \forall k, \forall q, \bar{x}_{qk} = \bar{x}_k$$

by variable, classes have the same means

Doesn't allow us to classify

$$\frac{\text{between inertia}}{\text{total inertia}} = 1 \Longrightarrow \forall k, \forall q, \forall i, x_{iqk} = \bar{x}_{qk}$$

individuals in the same class are identical

Ideal for classifying

It corresponds with 1 variable to the $\eta^2$ (or $R^2$ in analysis of variance) $\rightarrow$ percentage of variability explained by the partition (categorical variable).

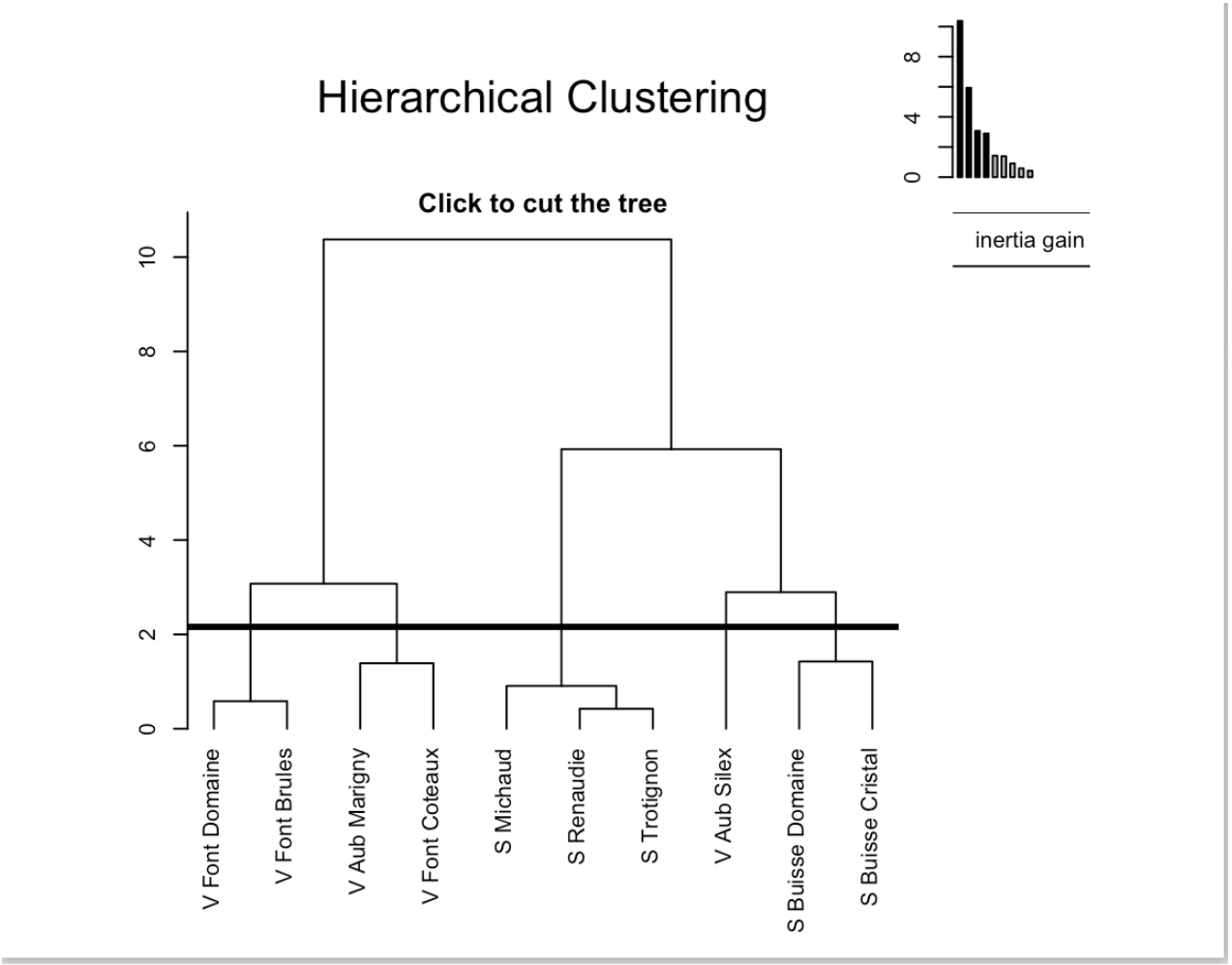
Warning: this criteria depends on the number of individuals and clusters so difficult to interpret

# Wine data

- 10 observations (rows): white wines from Val de Loire
- 27 variables (columns): sensory descriptors

| | O.fruity | O.passion | O.citrus | ... | Sweetness | Acidity | Bitterness | Astringency | Aroma.intensity | Aroma.persistency | Visual.intensity | Odor.preference | Overall.preference | Label |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| S Michaud | 4,3 | 2,4 | 5,7 | ... | 3,5 | 5,9 | 4,1 | 1,4 | 7,1 | 6,7 | 5,0 | 6,0 | 5,0 | Sauvignon |
| S Renaudie | 4,4 | 3,1 | 5,3 | ... | 3,3 | 6,8 | 3,8 | 2,3 | 7,2 | 6,6 | 3,4 | 5,4 | 5,5 | Sauvignon |
| S Trotignon | 5,1 | 4,0 | 5,3 | ... | 3,0 | 6,1 | 4,1 | 2,4 | 6,1 | 6,1 | 3,0 | 5,0 | 5,5 | Sauvignon |
| S Buisse Domaine | 4,3 | 2,4 | 3,6 | ... | 3,9 | 5,6 | 2,5 | 3,0 | 4,9 | 5,1 | 4,1 | 5,3 | 4,6 | Sauvignon |
| S Buisse Cristal | 5,6 | 3,1 | 3,5 | ... | 3,4 | 6,6 | 5,0 | 3,1 | 6,1 | 5,1 | 3,6 | 6,1 | 5,0 | Sauvignon |
| V Aub Silex | 3,9 | 0,7 | 3,3 | ... | 7,9 | 4,4 | 3,0 | 2,4 | 5,9 | 5,6 | 4,0 | 5,0 | 5,5 | Vouvray |
| V Aub Marigny | 2,1 | 0,7 | 1,0 | ... | 3,5 | 6,4 | 5,0 | 4,0 | 6,3 | 6,7 | 6,0 | 5,1 | 4,1 | Vouvray |
| V Font Domaine | 5,1 | 0,5 | 2,5 | ... | 3,0 | 5,7 | 4,0 | 2,5 | 6,7 | 6,3 | 6,4 | 4,4 | 5,1 | Vouvray |
| V Font Brûlés | 5,1 | 0,8 | 3,8 | ... | 3,9 | 5,4 | 4,0 | 3,1 | 7,0 | 6,1 | 7,4 | 4,4 | 6,4 | Vouvray |
| V Font Coteaux | 4,1 | 0,9 | 2,7 | ... | 3,8 | 5,1 | 4,3 | 4,3 | 7,3 | 6,6 | 6,3 | 6,0 | 5,7 | Vouvray |

# Hierarchical clustering of wine

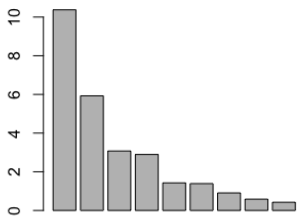


# Selecting the number of clusters

- Using the tree, depends of the use - the overall appearance of the tree
- The number of clusters, which must not be too high so as not to impede the concise nature of the approach.
- Interpretability of the clusters
- Inertia barplot

In FactoMineR: suggests a division into $Q$ clusters when the increase of between-clusters inertia when going from a $Q+1$ to a $Q$ clusters partition is much greater than that from a $Q$ to a $Q+1$ clusters partition.

# Inertia plot




Loss in between-inertia when going from

- 2 clusters to 1: 10.37
- 3 clusters to 2: 5.92
- 4 clusters to 3: 3.07
- ...

Important loss when going from 4 clusters to 3 clusters thus we prefer to keep 4 clusters

```
 res.hcpc$call$t$inert.gain
[1] 10.3743969  5.9271832  3.0749287  2.8940507  1.4259050
```

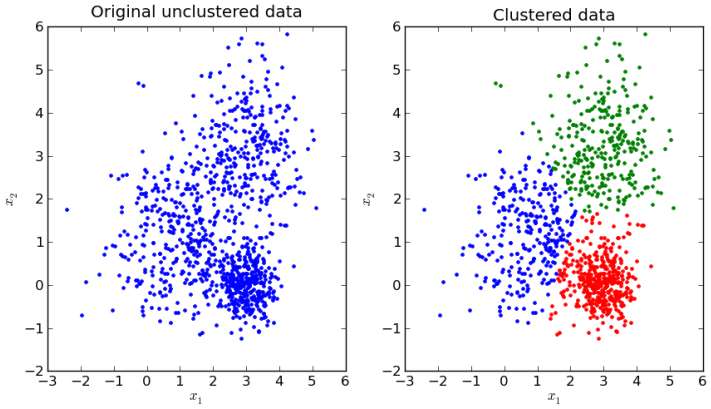
# K-means clustering

# Partition method: K-Means

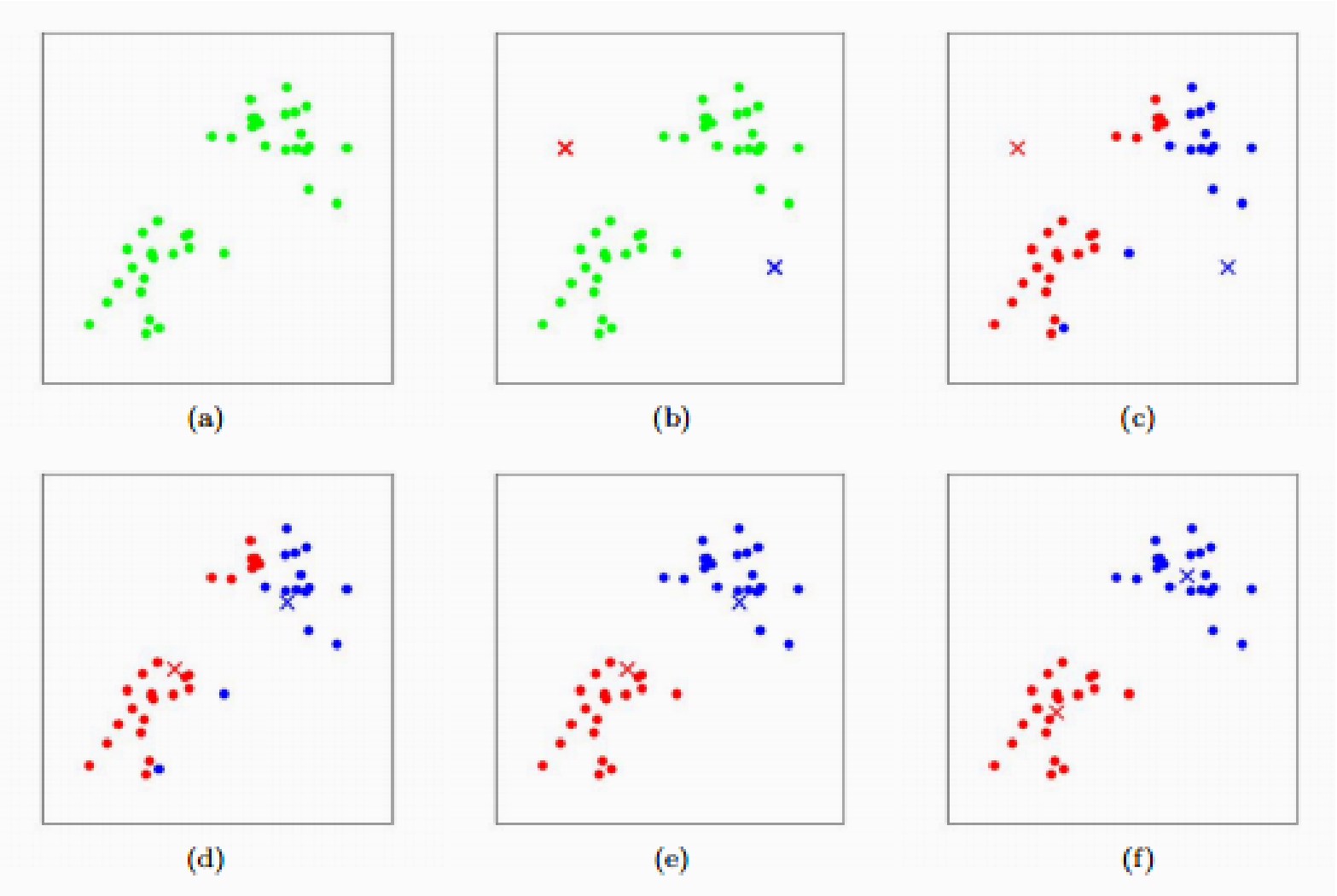
Simple and elegant approach for partitioning a data set into Q non-overlapping clusters.

To perform K-means:

- Specify the desired number of clusters Q.
- Similar: small Euclidian distance
- Objective: a good clustering is one for which the within-cluster variation is as small as possible.



# kmeans: illustration



# Objective

- Fix $Q \geq 2$, $I$ data points $x_{i.} \in \mathbb{R}^K$
- Find partition $C_1, \ldots, C_Q$ such that minimizes the quantification error

$$\sum_{q=1,\ldots,Q} \sum_{i \in C_q} \sum_{k=1}^{K} (x_{ik} - \bar{x}_{qk})^2$$

  $\Rightarrow$ We want to partition the observations into $Q$ clusters such that the total within-cluster variation, summed over all $Q$ clusters, is as small as possible.
- Impossible to find the exact solution!

# kmeans algorithm

Lloyd (1981) proposes a way of finding local solutions

- Choose at random $Q$ centroids $\{c_1, \ldots, c_Q\}$ K clusters
- For each $q \in \{1, \ldots, Q\}$, find the set $C_q$ of points that are closer to $m_q$ than any $m_{q'}$ for $q' \neq q$
- Update the centroids (with means, hence the name)

$$m_{q.} = \frac{1}{|C_q|} \sum_{i \in C_q} x_{i.}$$

- Repeat the two previous steps until the sets $C_q$ don't change

# Conclusion

In order to perform clustering, some decisions must be made:

- Should the features first be standardized?
- In case of hierarchical clustering:
  - What dissimilarity measure should be used?
  - What type of linkage should be used?
  - Where should we cut the dendogram in order to obtain clusters?
- In case of K-means clustering:
  - How many clusters should we look for the data?
- In practice, we try several different choices, and look for the one with the most useful or interpretable solution.

There is no single right answer!

# Conclusion - Comparison

- AHC do not require numbers of clusters (shape of the tree)
- Users usually like hierarchical clusters
- Assumption of hierarchical structure might be unrealistic? (Ex a group of people with a 50/50 split of males and females, evenly split among Americans, Japanese, and French. Imagine a scenario in which the best division into 2 groups might split by gender, and the best division into 3 groups might split them by nationality. In this case, the true clusters are not nested, i.e. the best division into 3 groups does not result from taking the best division into 2 groups and splitting up one of those groups. Consequently, this situation could not be well represented by hierarchical clustering.)
- AHC slow, difficult for large sample size
- K-means requires the choice of $Q$
- Fast.
- Dependent on the initialisation

# Be aware...

$\Rightarrow$ Any time clustering is performed on a data set we will find clusters!!

- Represent true groups or clustering noise?
- Is my clustering stable? : performing clustering on subsamples with different choices of the parameters $\rightarrow$ what patterns consistently emerge

$\Rightarrow$ Be careful about how the results of a clustering analysis are reported. (Susan Holmes: Reproducible research workflow in R for the analysis of personalized human microbiome data - NIPS Talk)

$\Rightarrow$ These results should not be taken as the absolute truth about a data set, rather, they should constitute a starting point for the developments of a scientific hypothesis and further study.

# Clustering in practice

- Function HCPC in FactoMineR
- Function kmeans in R

# Complementarity between PCA, AHC and K-means

# K-means after hierarchical clustering

The partition obtained by hierarchical clustering is not optimal and can be improved using K-means

Algorithm:

- use the obtained hierarchical partition to initialize K-means
- run a few iterations of kmeans

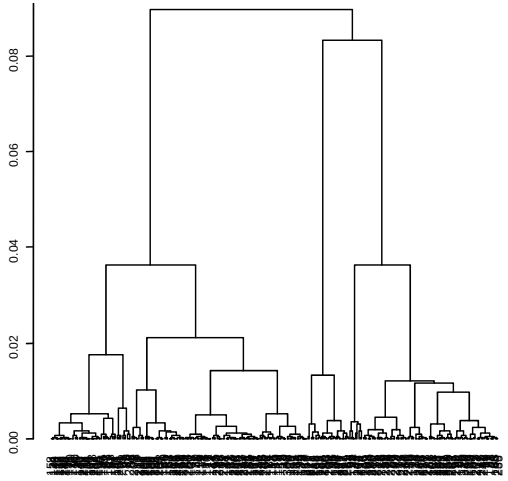
$\Longrightarrow$ potentially improved partition

Advantage: more "robust" partition

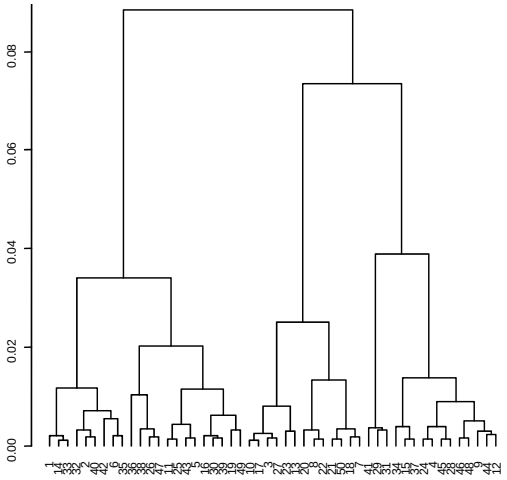
Disadvantage: loss of hierarchical structure

# Hierarchical clustering in high dimension

- If many individuals, hierarchical algorithm is too long
  - Use K-means to partition into around 100 classes
  - Build tree using these classes (weighted by the number of individuals in each class)
  - Idea: gives us the "top" of the tree




Tree from original data




Tree using classes

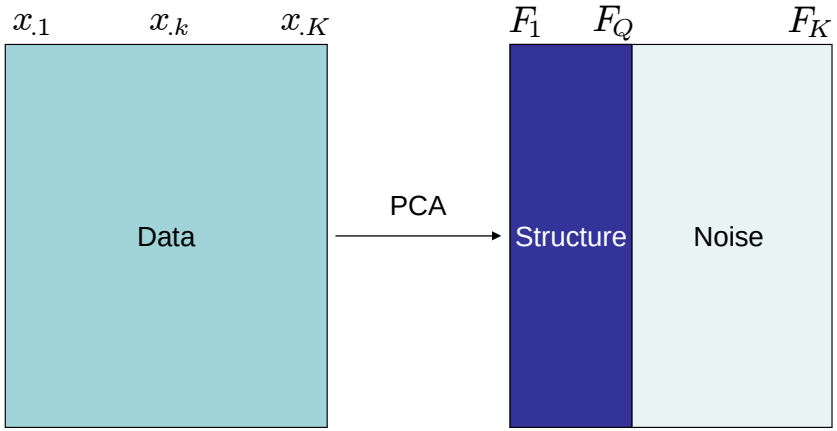
- If many variables: do PCA and keep only first dimensions $\Longrightarrow$ takes us to classical case

# PCA before clustering (pre-processing)

$\Rightarrow$ AHC or K-means on the data

$\Rightarrow$ AHC or K-means on the principal components

PCA transforms the raw variables into orthogonal principal components $F_{.1}, ..., F_{.Q}$ with decreasing variance $\lambda_1 \geq \lambda_2 \geq ...\lambda_Q$

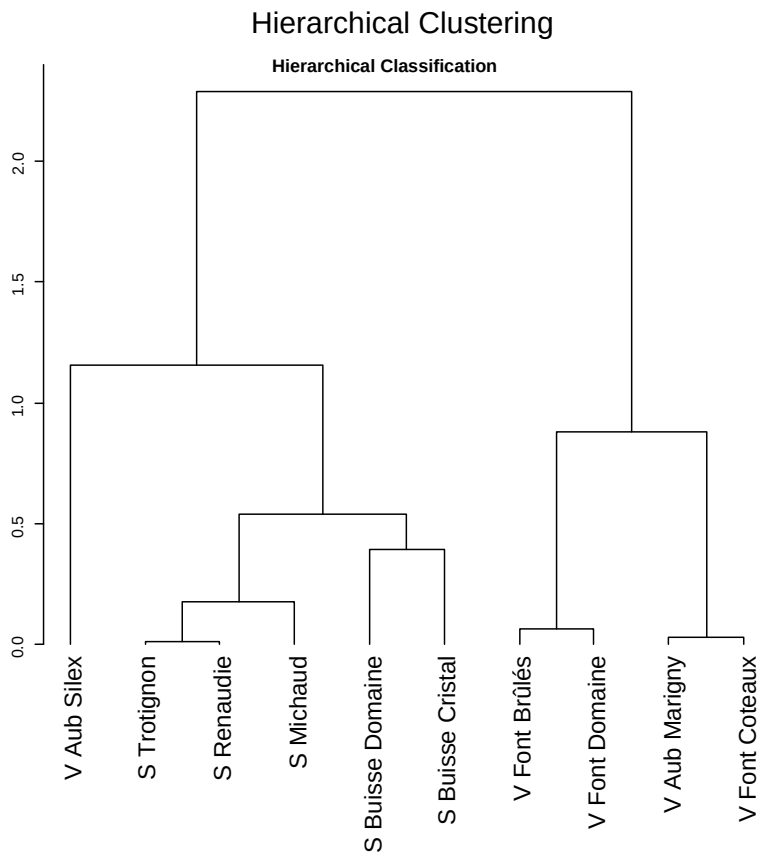



$\Rightarrow$ Keeping the first components makes the clustering more stable (denoise)

$\Rightarrow$ But how many components do we keep? 95% of inertia

# Representation of Hierarchical clustering using PCA

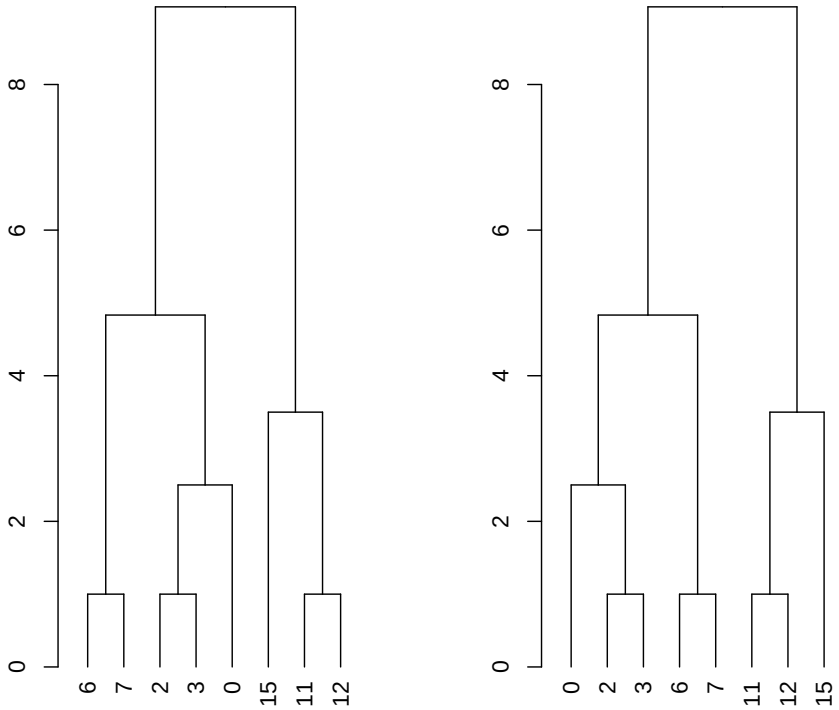
AHC onto the first 5 principal components



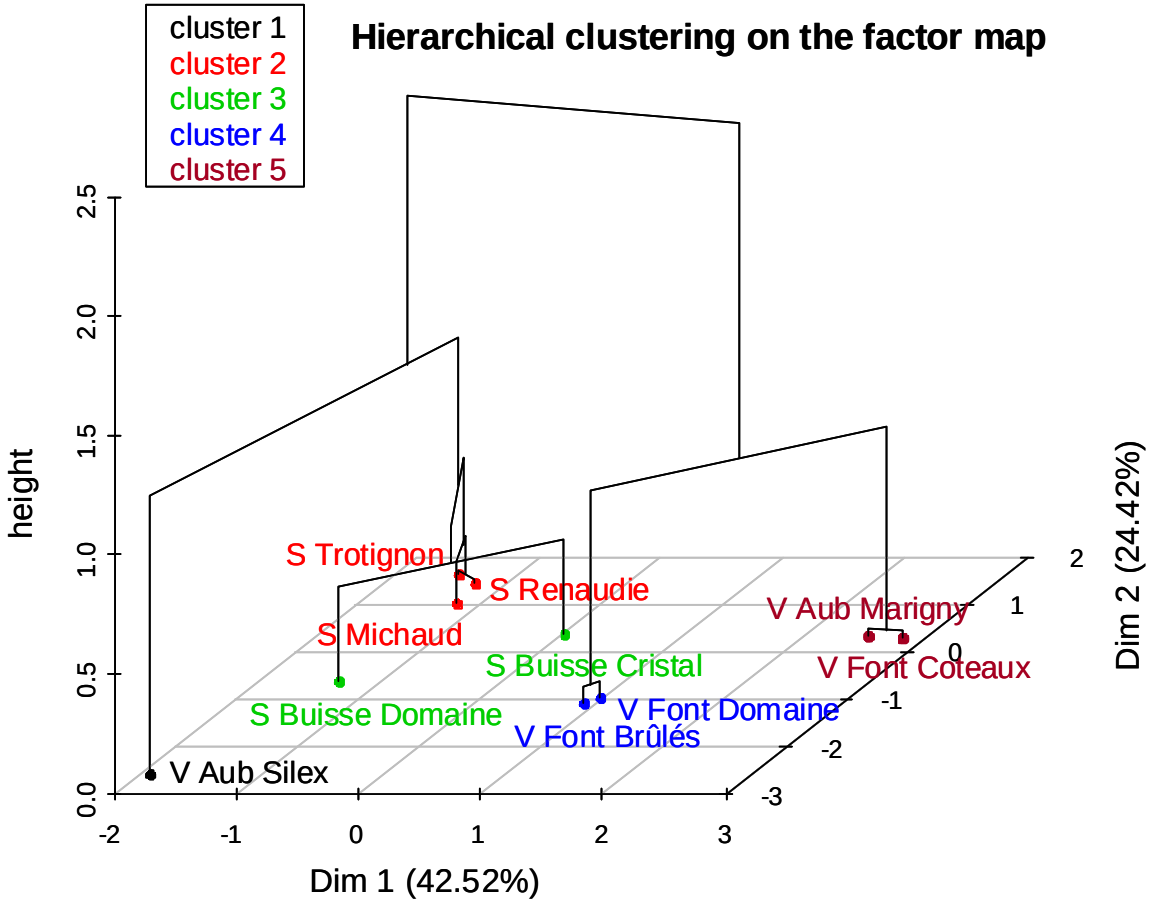

observations are sorted according to their coordinate on $F_{.1}$

# Why sorting the tree?

```
X <- c(6,7,2,0,3,15,11,12)
names(X) <- X
library(cluster)
par(mfrow=c(1,2))
plot(as.dendrogram(agnes(X)))
plot(as.dendrogram(agnes(sort(X))))
```

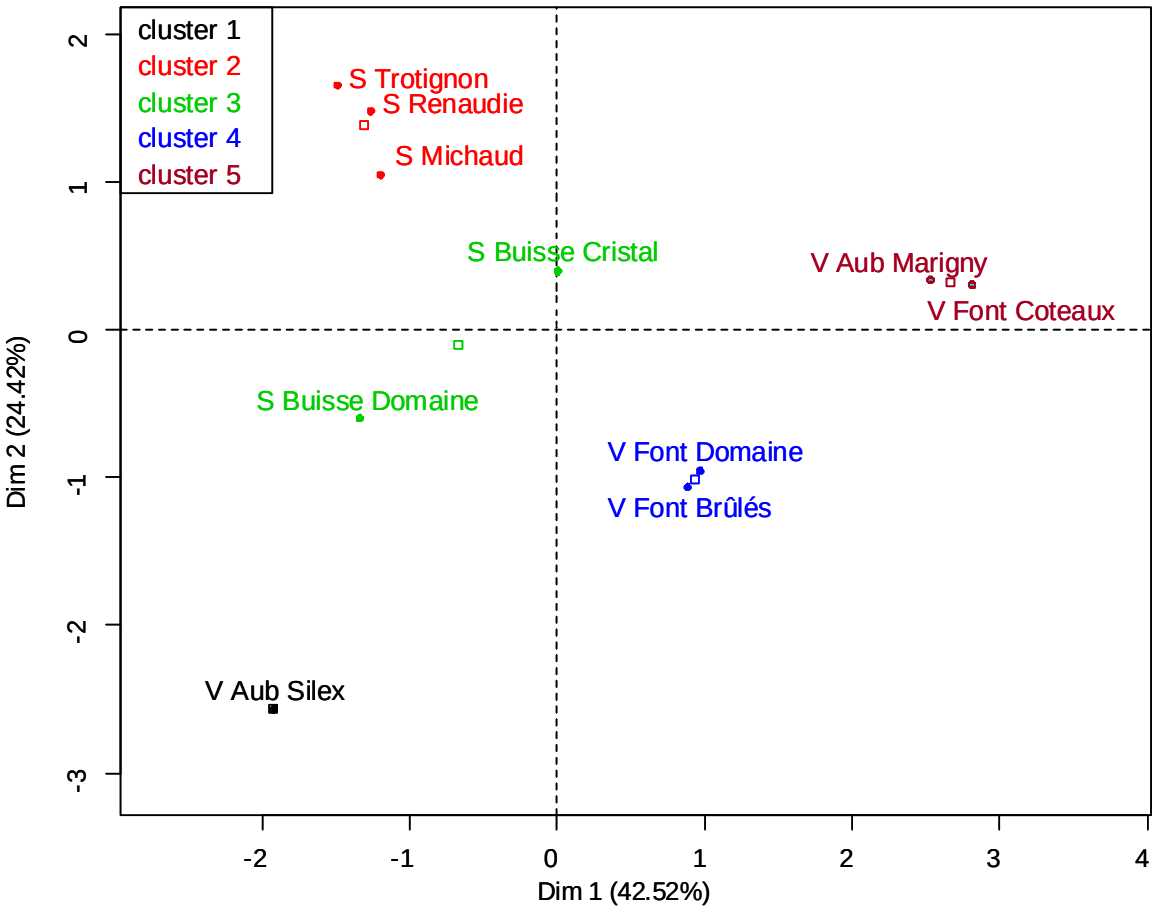


# Hierarchical tree on the principal components map




Hierarchical tree gives an idea of the other dimensions

# Partition on the principal components map




Continuous vision (principal components) and discontinuous (clusters)

# What to do with categorical data???

**Which distance to use?**

- two categories: Jaccard index, Dice's coefficient simple match, etc. Indices well- fitted for presence/absence data
- with more than 2 categories: use for example the $\chi^2$ distance

Counterpart of PCA for categorical data: Multiple Correspondence Analysis

**MCA as a preprocessing**

- transform categorical variables in continuous ones
- delete the last dimensions to denoise (more stable clustering)
- clustering on all MCA dimensions: clustering with $\chi^2$ distances

# Multiple Correspondence Analysis (MCA)

$X_{I \times m}$ $m$ categorical variables coded with indicator matrix $A$

$$X = \begin{array}{|c|c|c|} \hline y & \dots & attack \\ y & \dots & attack \\ y & \dots & attack \\ n & \dots & suicide \\ & & \\ & & \\ & & \\ n & \dots & accident \\ n & \dots & suicide \\ \hline \end{array} \quad A = \begin{array}{|cc|c|ccc|} \hline 1 & 0 & \dots & 1 & 0 & 0 \\ 1 & 0 & \dots & 1 & 0 & 0 \\ 1 & 0 & \dots & 1 & 0 & 0 \\ 0 & 1 & \dots & 0 & 1 & 0 \\ & & & & & \\ & & & & & \\ & & & & & \\ 0 & 1 & \dots & 0 & 0 & 1 \\ 0 & 1 & \dots & 0 & 1 & 0 \\ \hline \end{array} \quad D_p = \begin{array}{|ccc|} \hline p_1 & & 0 \\ & \ddots & \\ 0 & & p_J \\ \hline \end{array}$$

For a category $c$, the frequency of the category: $p_c = I_c / I$.
A SVD on weighted matrix: $Z = \frac{1}{\sqrt{mI}}(A - 1p^T)D_p^{-1/2} = U \Lambda V'$

The PC ($F = U\Lambda$) satisfies: $F_l = \arg\max_{F_l \in \mathbb{R}^I} \quad \frac{1}{m}\sum_{j=1}^{m} \eta^2(F_l, X_j)$

$\eta^2(F_l, X_j) = \frac{\sum_{c=1}^{C_j}(F_{.c} - F_{..})^2}{\sum_i \sum_c (F_{ic})^2}$

Benzecri, 1973 : *"All in all, doing a data analysis, in good mathematics, reduces to computing eigenvectors; all the science (or the art) of it is in finding the right matrix to diagonalize"*